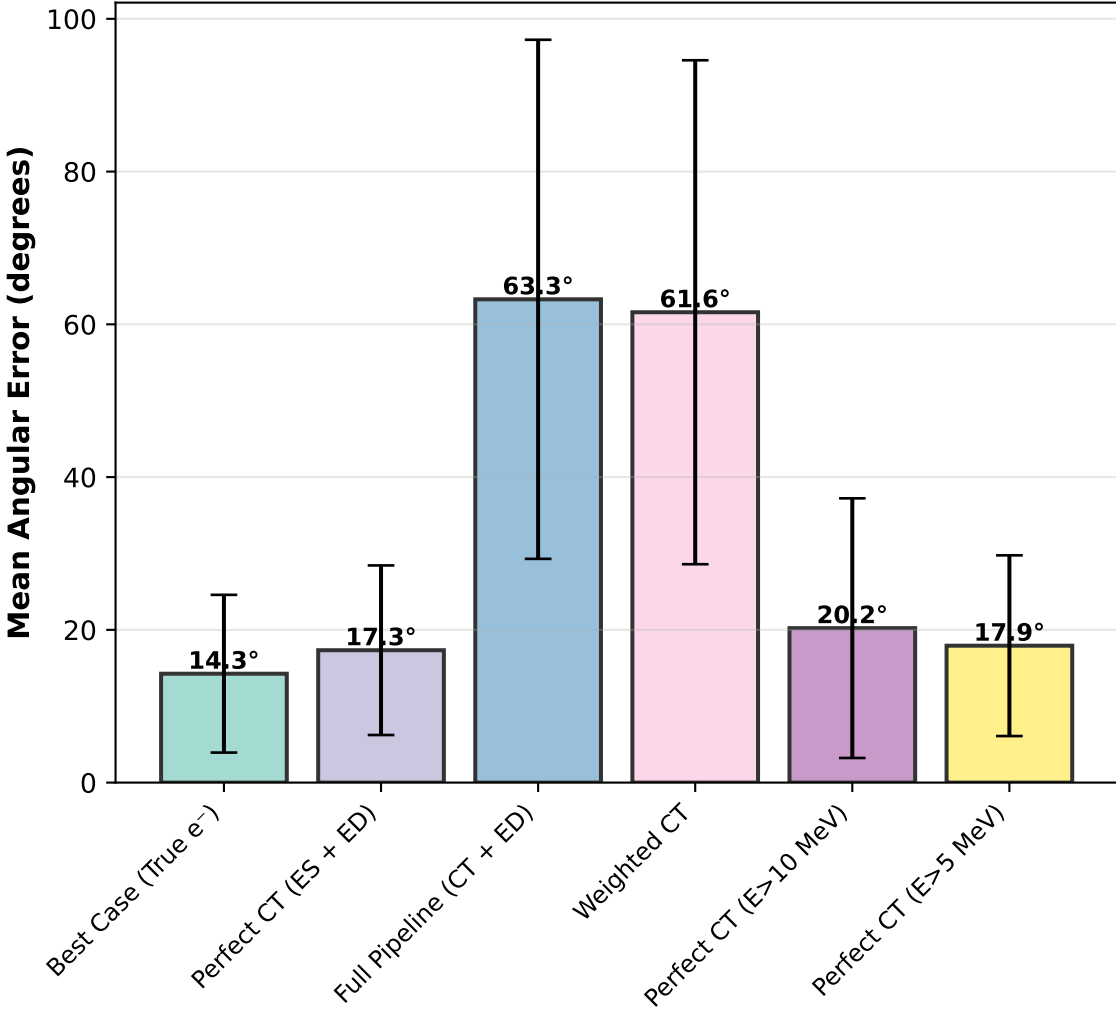


Emcee 6-Scenario Aggregate Analysis

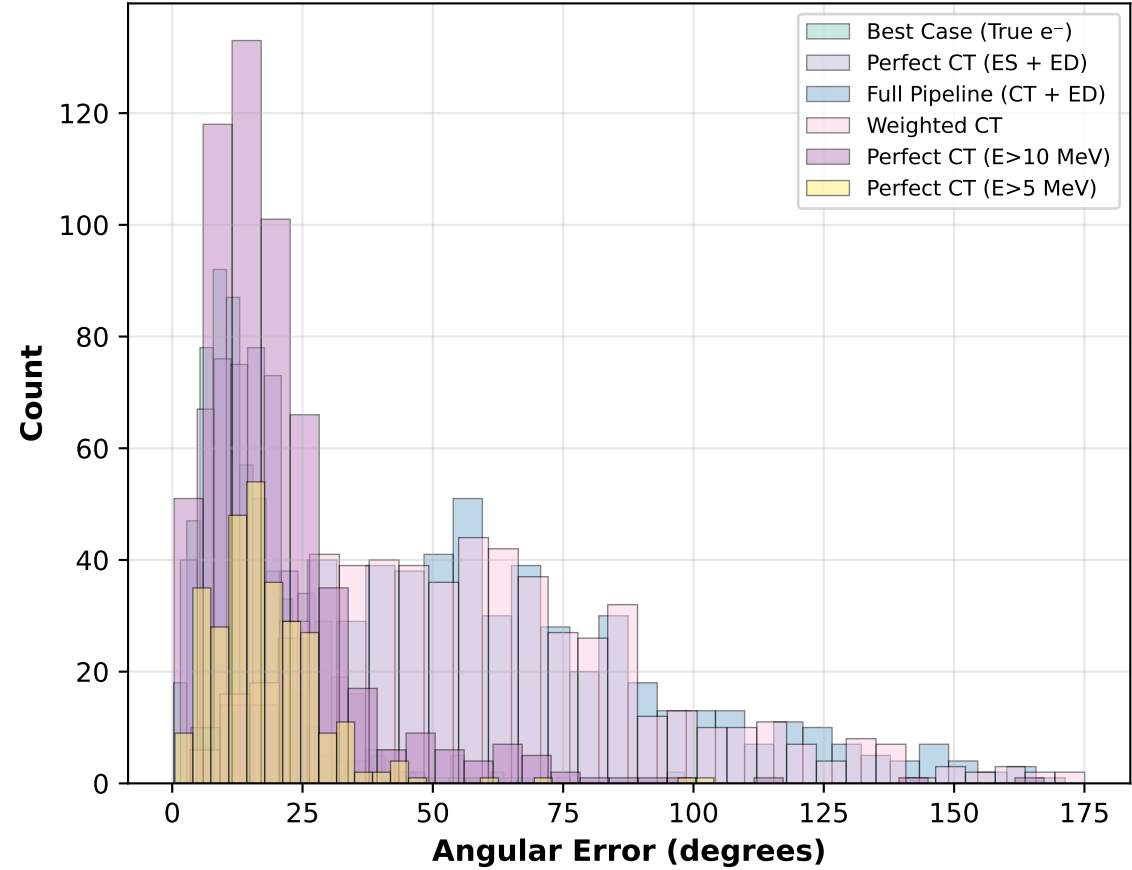
Summary Statistics

Scenario	N	68% Q	Median	Mean
Best Case (True e ⁻)	567	15.78°	11.81°	14.26°
Perfect CT (ES + ED)	567	19.69°	15.98°	17.33°
Full Pipeline (CT + ED)	567	73.93°	56.62°	63.28°
Weighted CT	567	71.97°	57.73°	61.58°
Perfect CT (E>10 MeV)	567	21.34°	16.29°	20.22°
Perfect CT (E>5 MeV)	299	20.36°	16.15°	17.93°

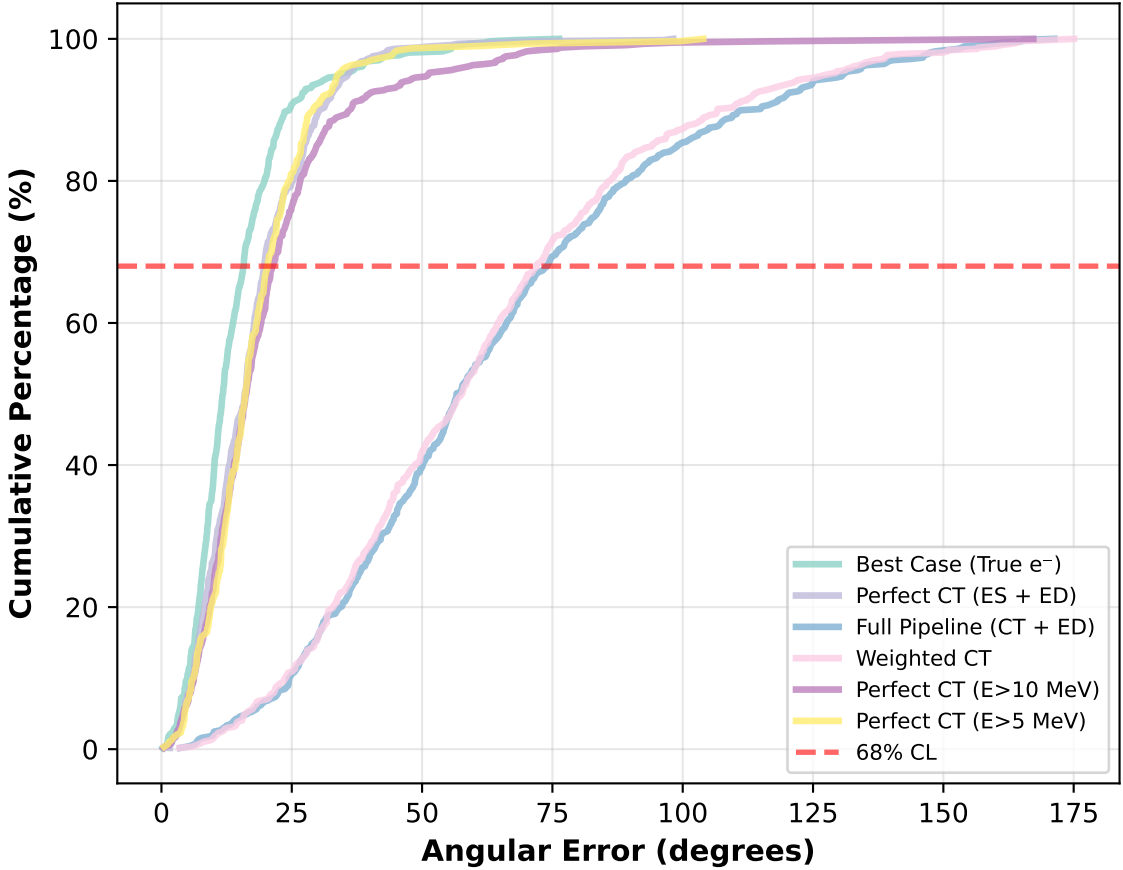
Mean Resolution ± Std Dev



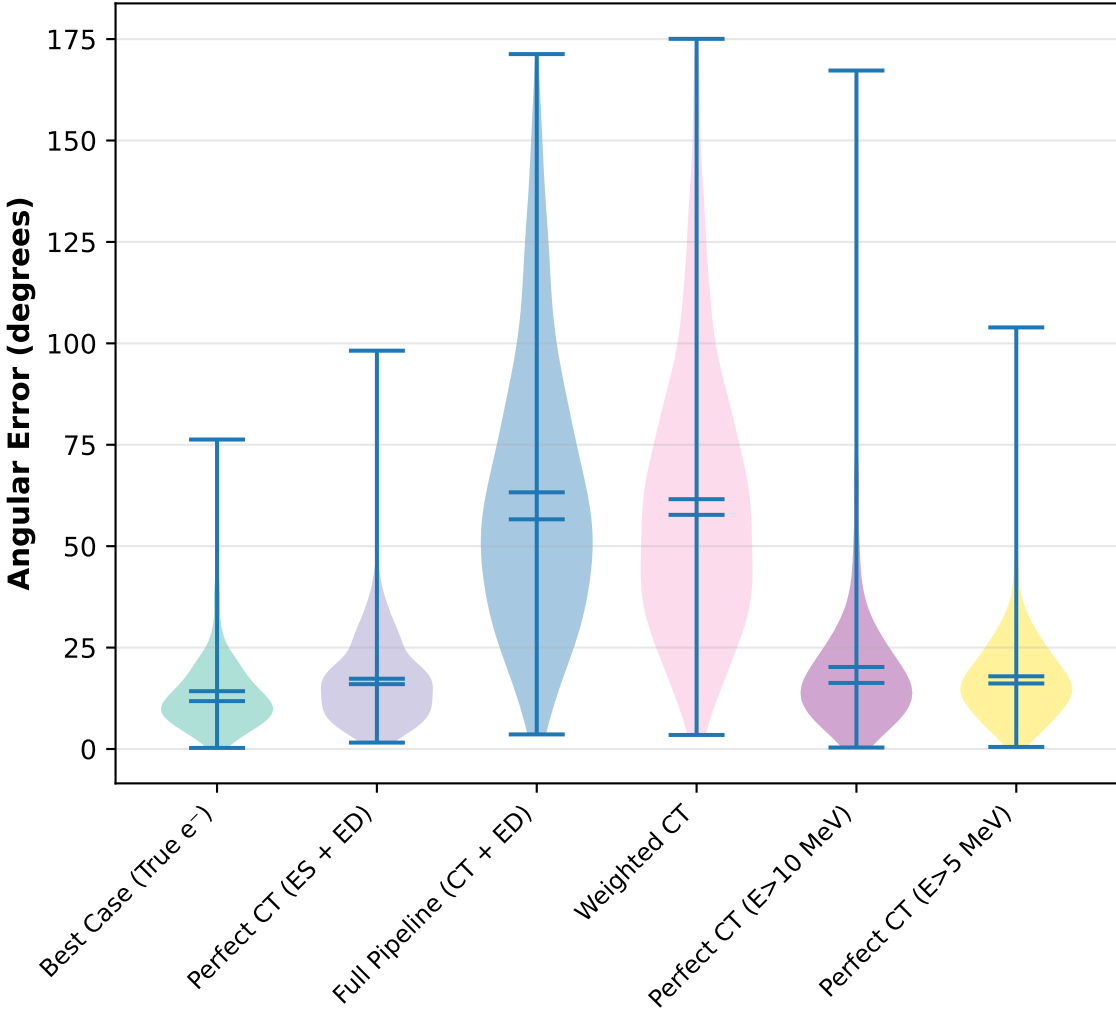
Distribution Overlay



Cumulative Distribution

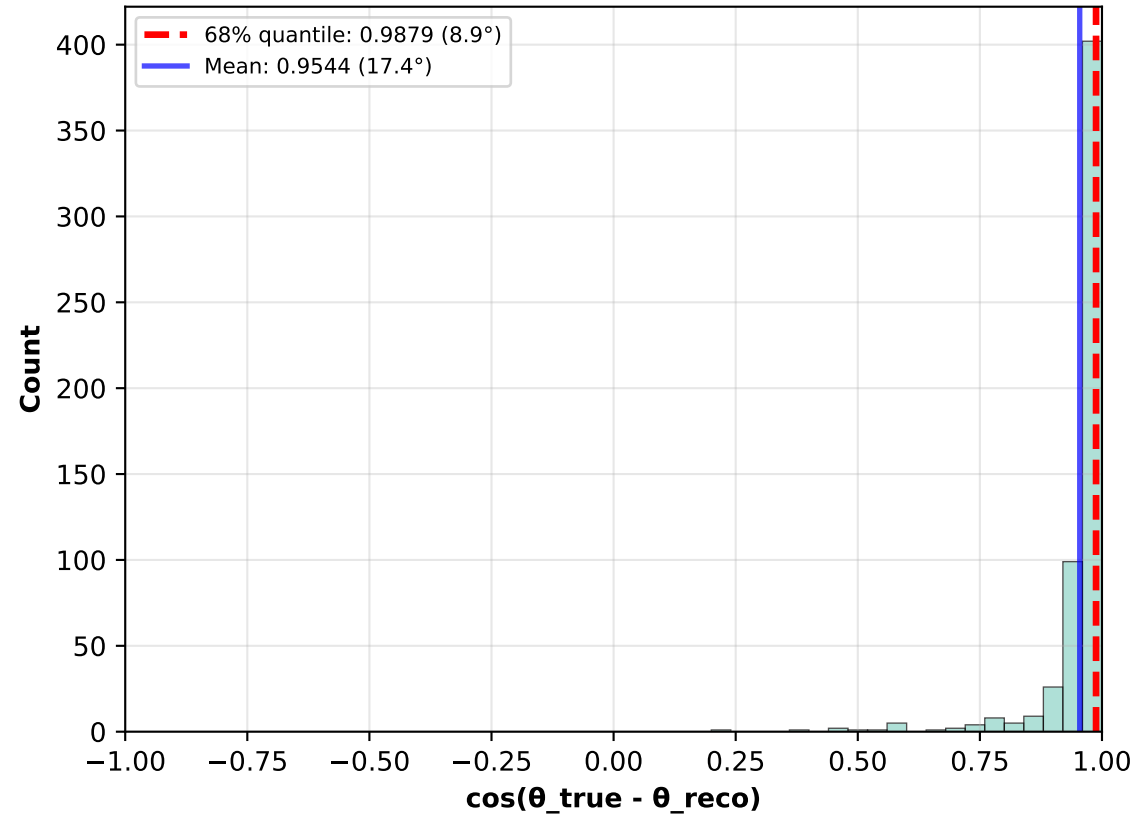


Distribution Shapes

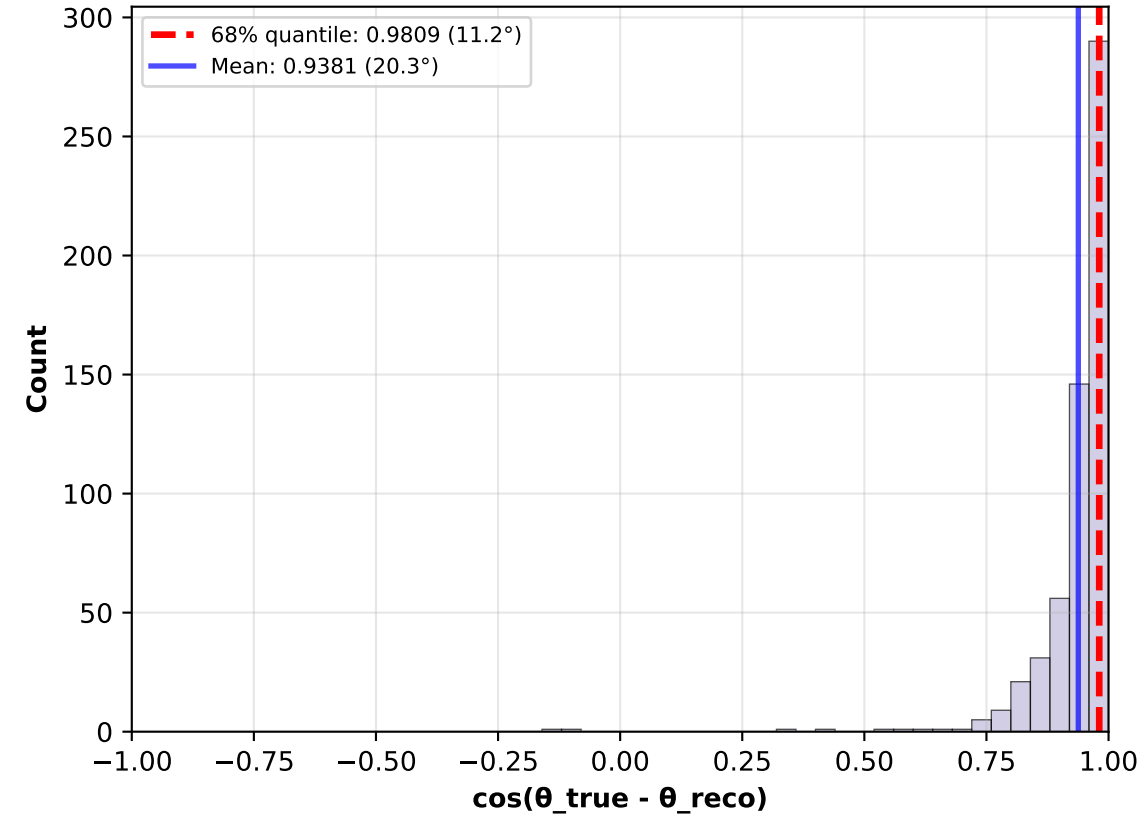


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Full Range

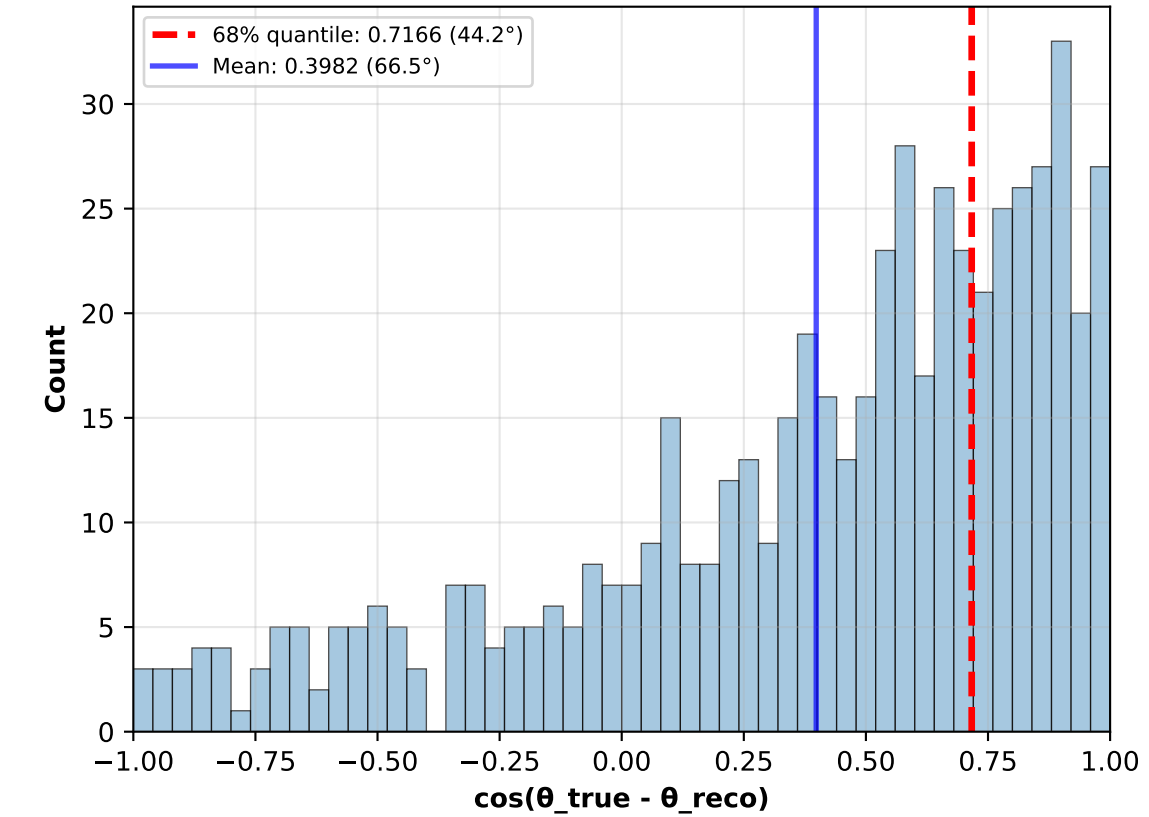
Best Case (True e^-)
(N=567)



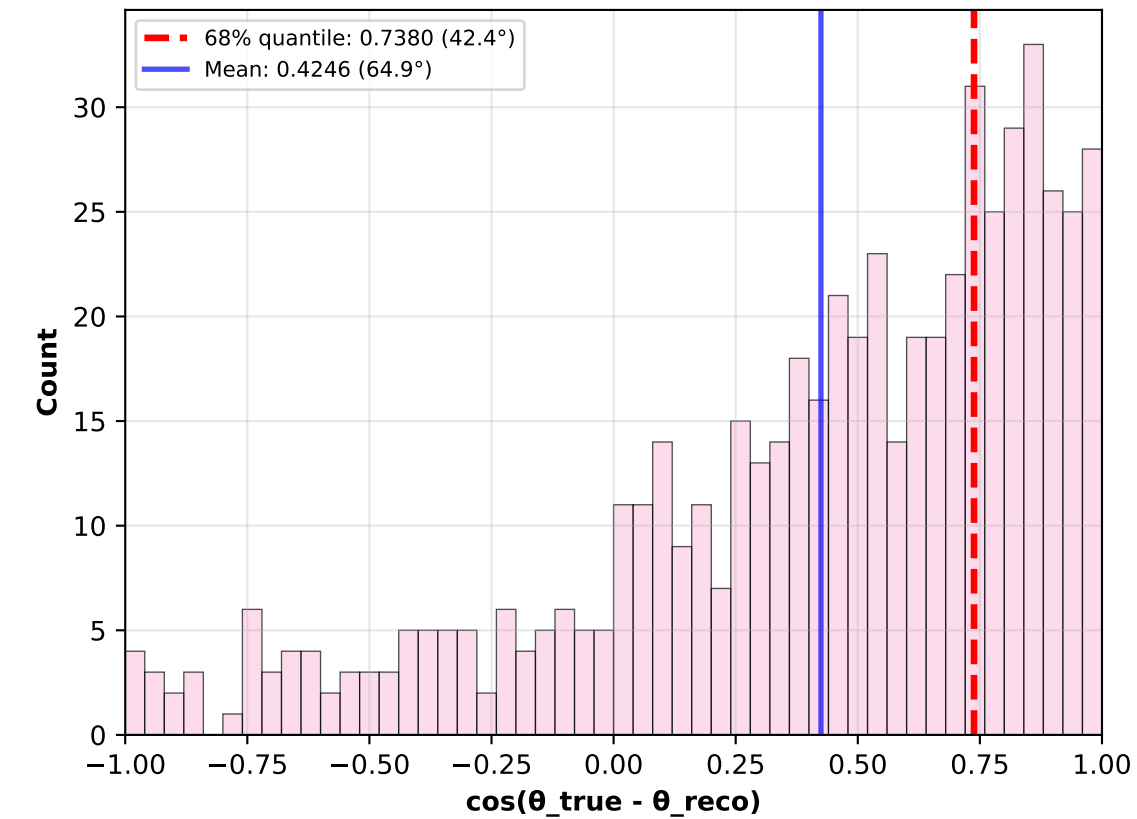
Perfect CT (ES + ED)
(N=567)



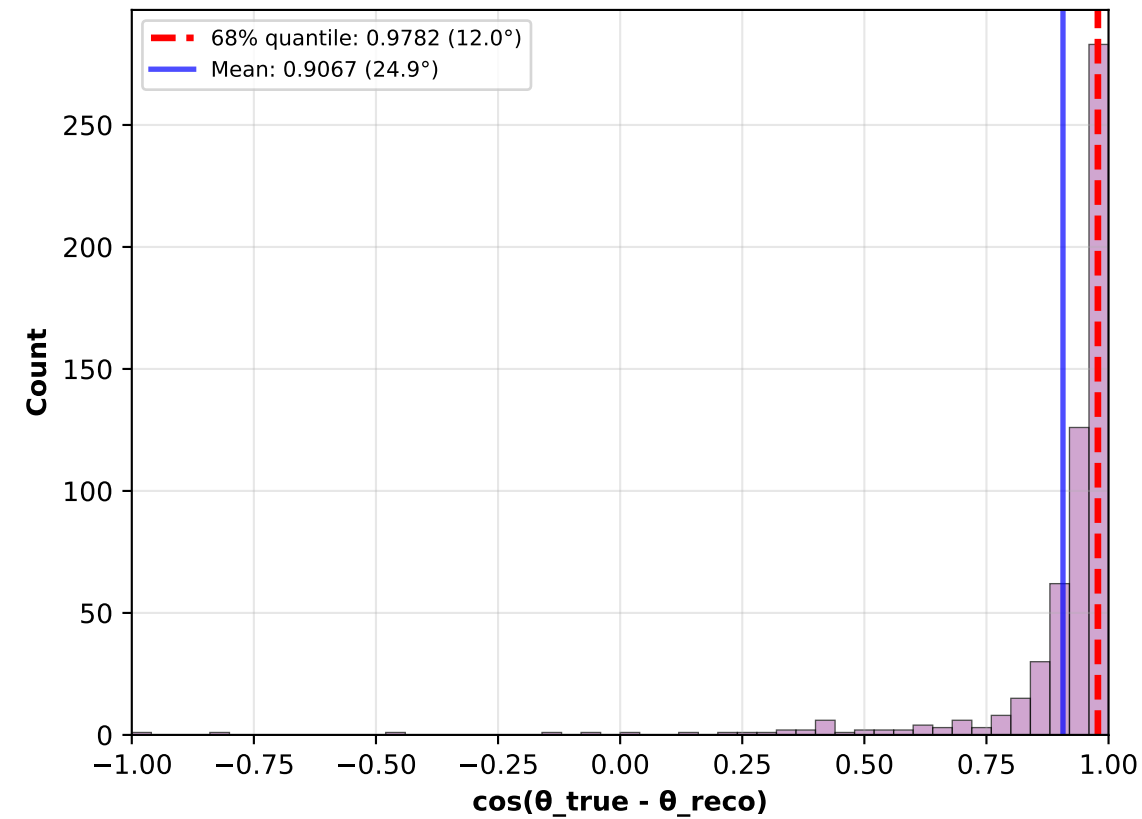
Full Pipeline (CT + ED)
(N=567)



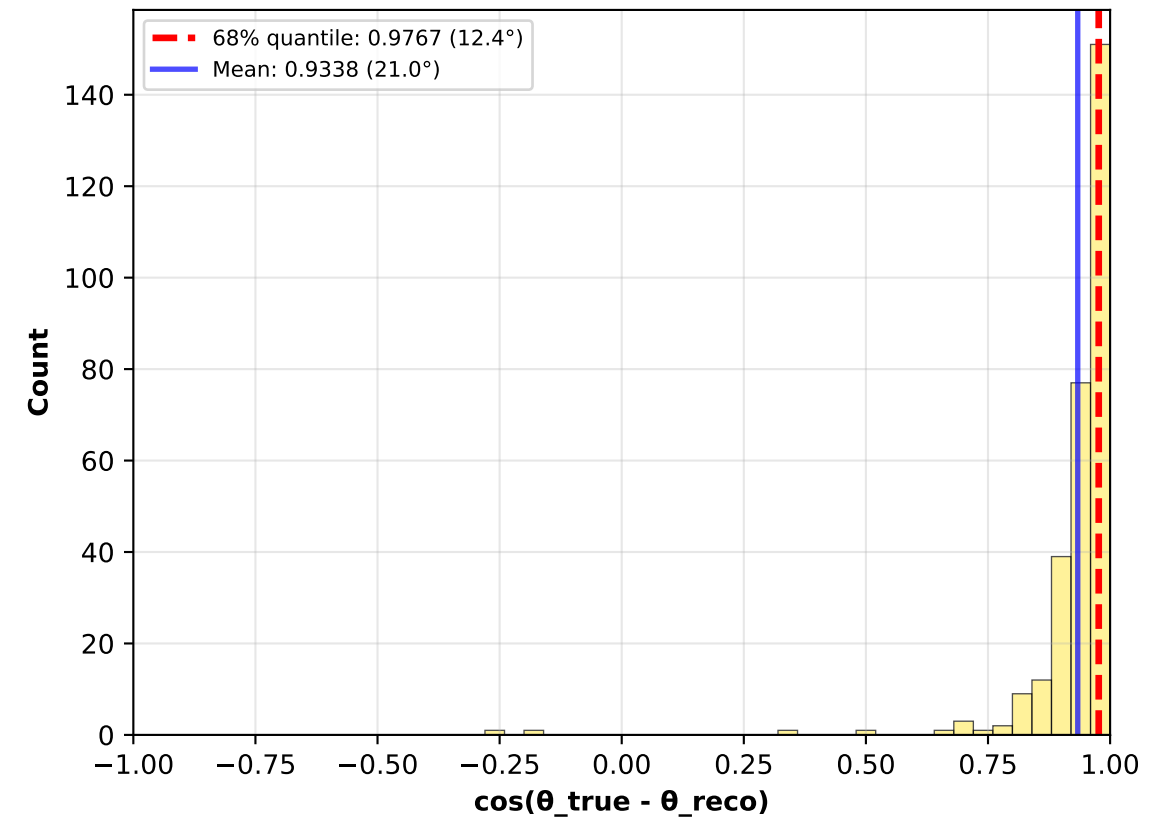
Weighted CT
(N=567)



Perfect CT (E>10 MeV)
(N=567)

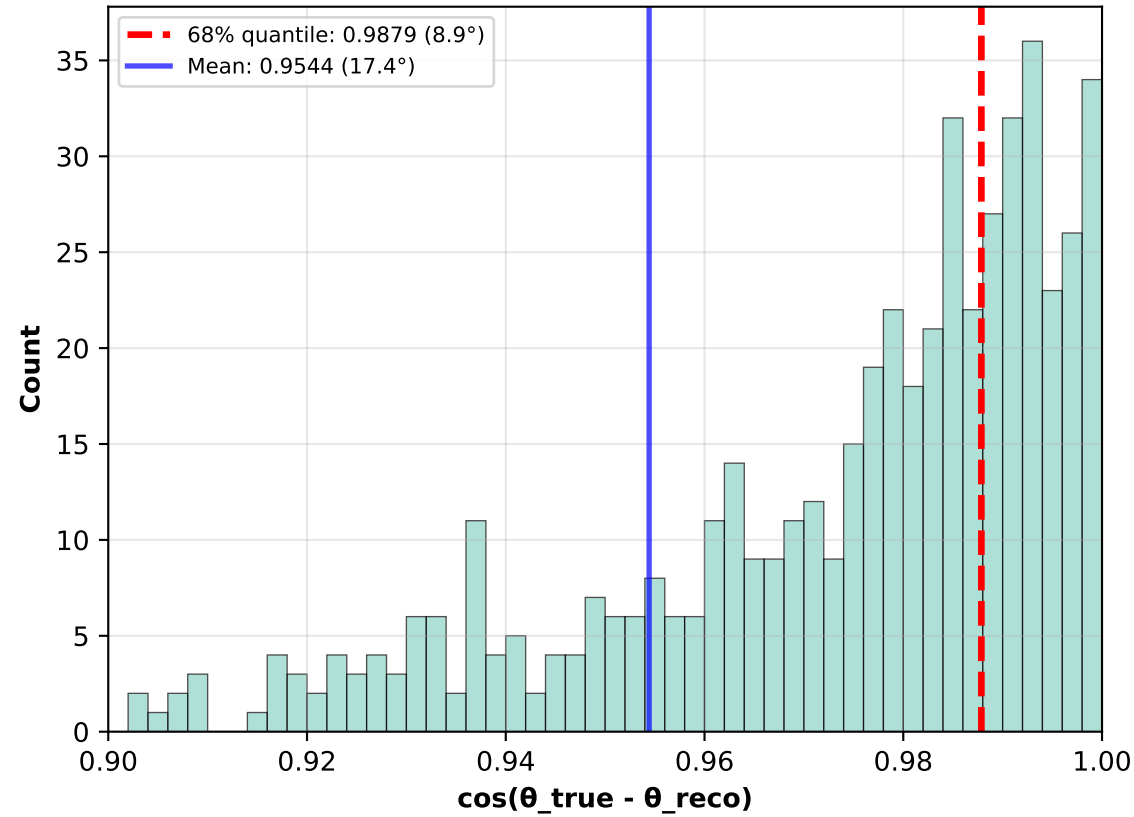


Perfect CT (E>5 MeV)
(N=299)

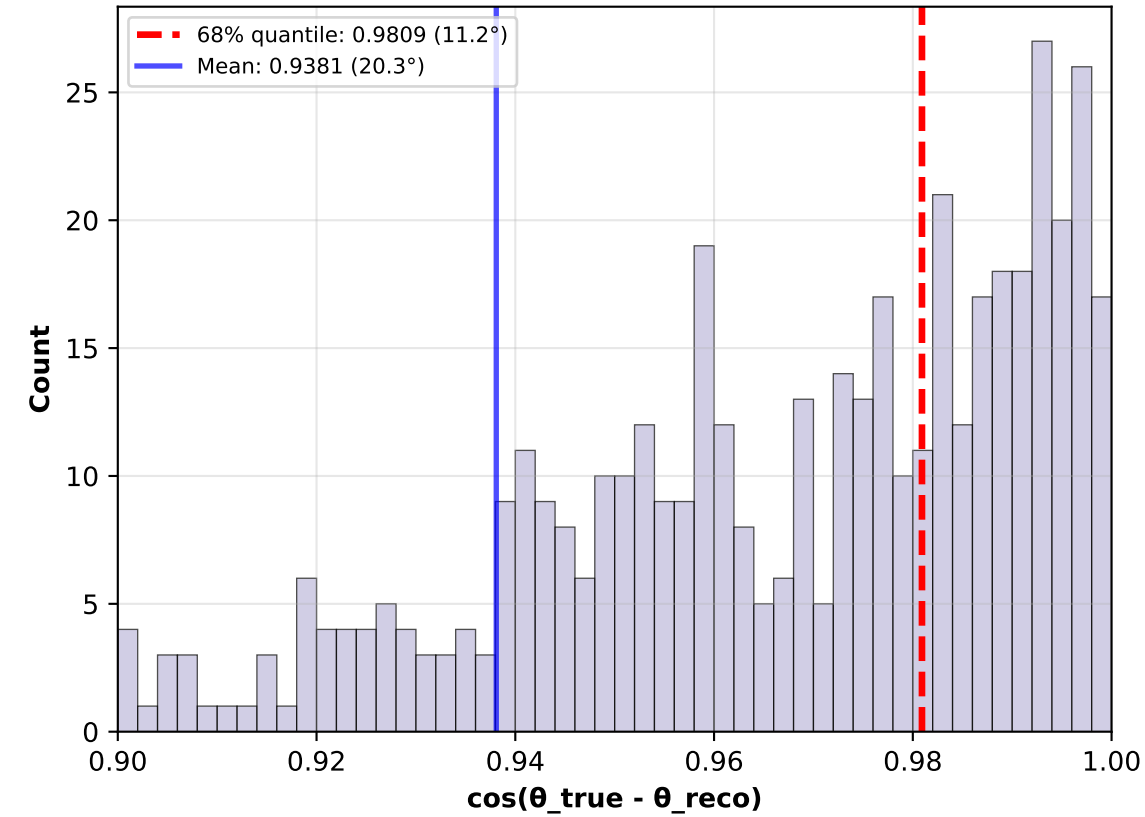


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Zoomed (0.9-1.0)

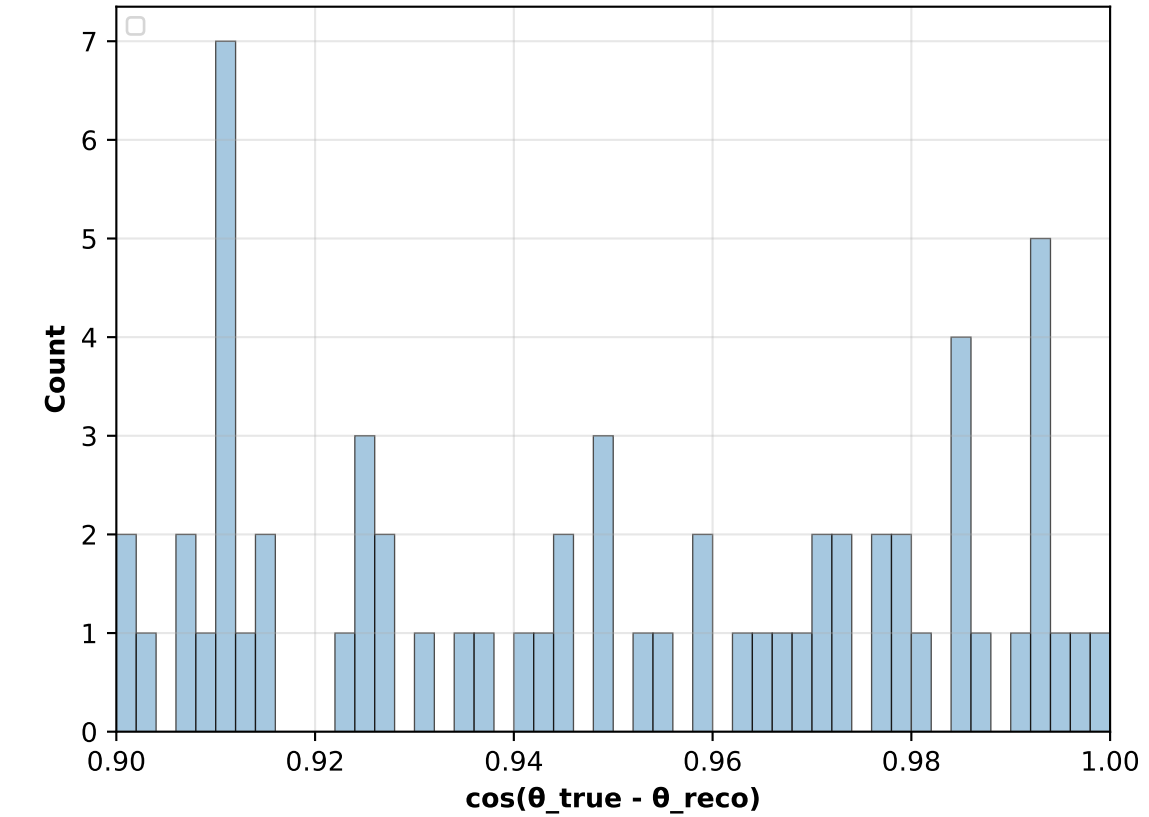
Best Case (True e^-)
(N=567)



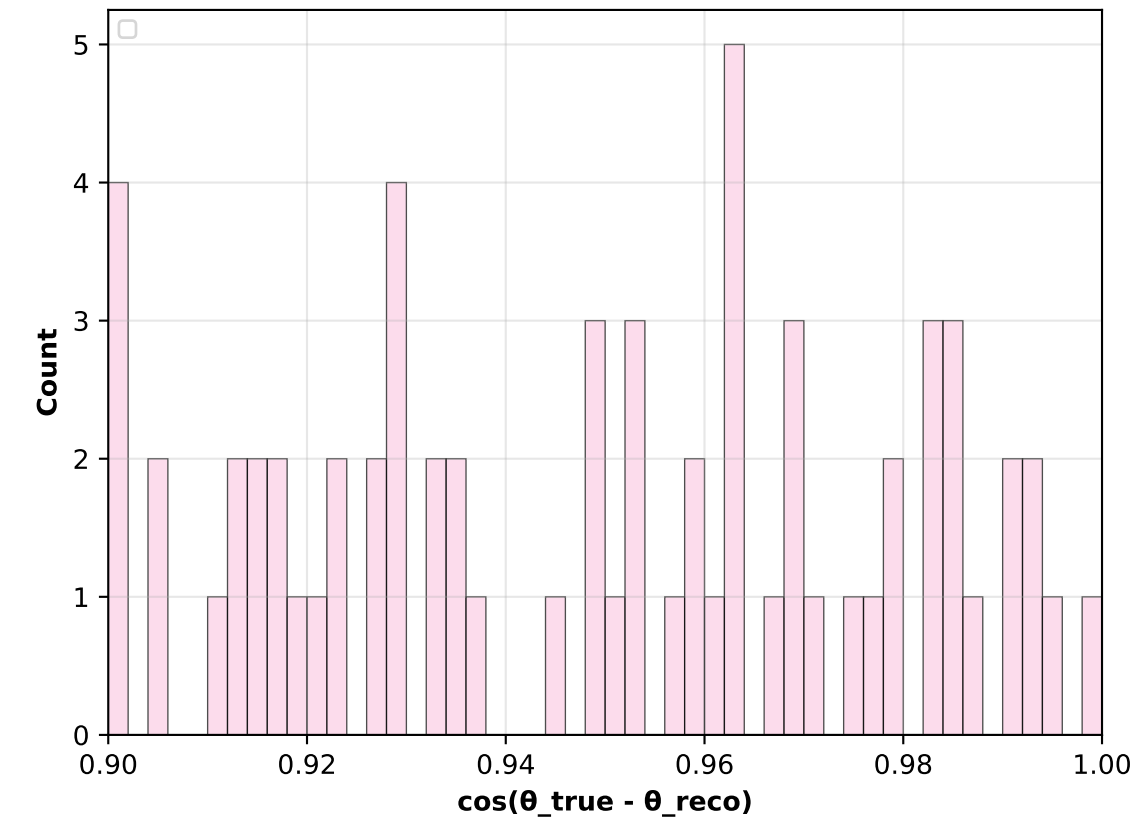
Perfect CT (ES + ED)
(N=567)



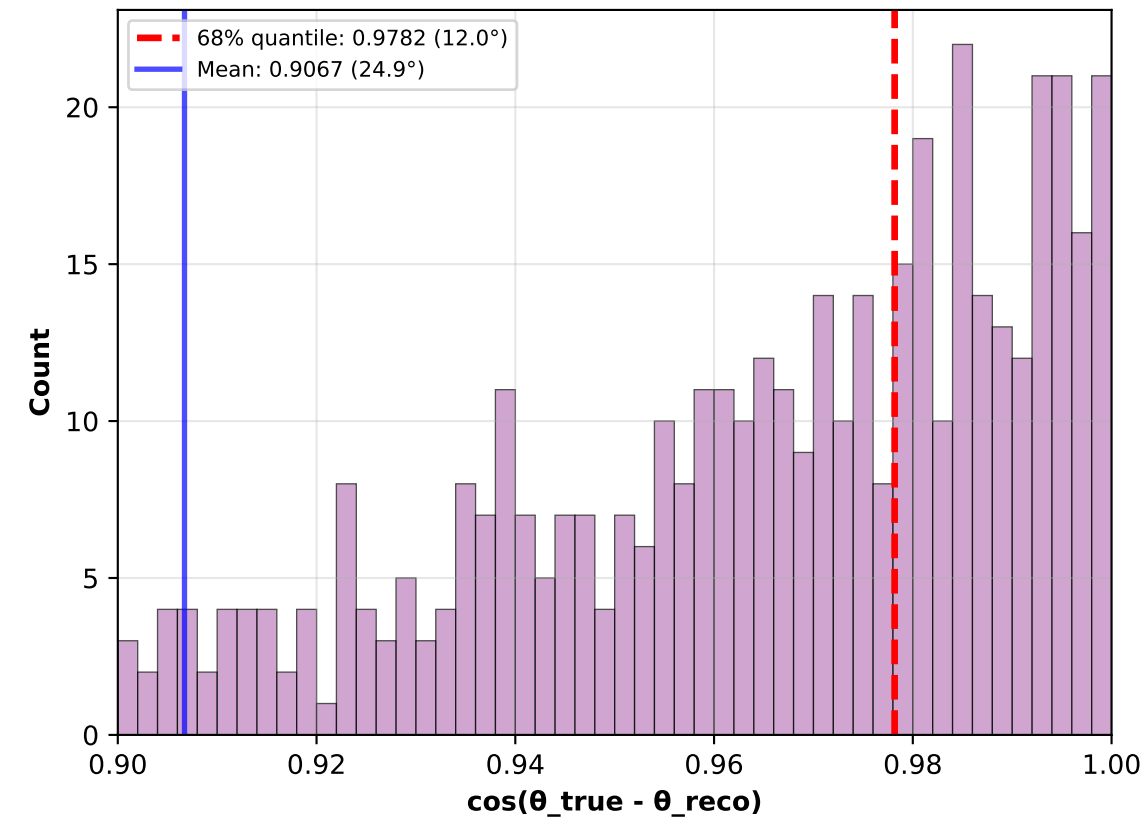
Full Pipeline (CT + ED)
(N=567)



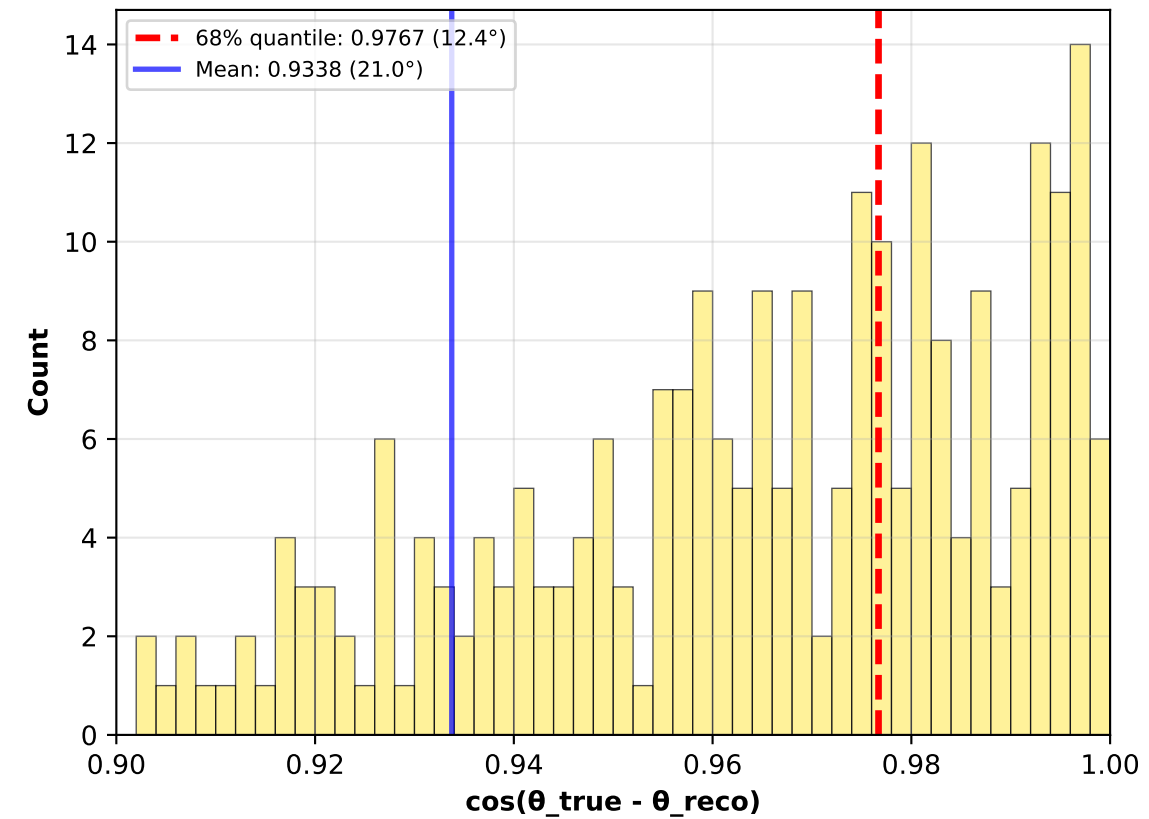
Weighted CT
(N=567)



Perfect CT (E>10 MeV)
(N=567)



Perfect CT (E>5 MeV)
(N=299)



Cluster Usage Across Scenarios

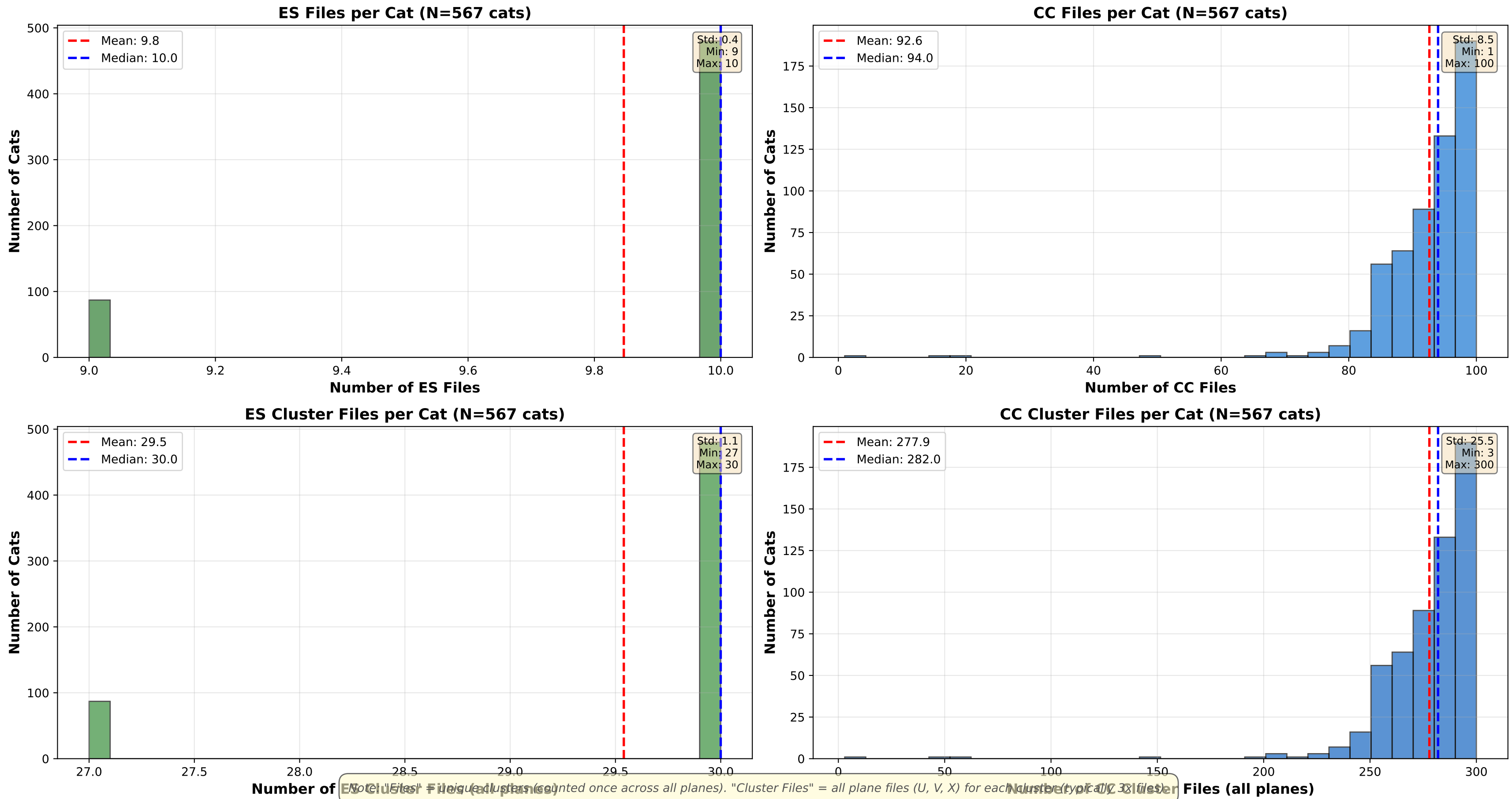
Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
Best Case (True e ⁻)	567	102	10	271	MCMC samples (weighted sum)
Perfect CT (ES + ED)	567	102	10	271	MCMC samples (weighted sum)
Full Pipeline (CT + ED)	567	102	10	3243	MCMC samples (weighted sum)
Weighted CT	567	102	10	3243	MCMC samples (weighted sum)
Perfect CT (E>10 MeV)	567	102	10	147	143.6% retention
Perfect CT (E>5 MeV)	299	101	10	242	MCMC samples (weighted sum)

Note: "Clusters Used" shows different meanings per scenario:

- Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
- Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
- Energy cut scenarios: Number of clusters passing energy threshold

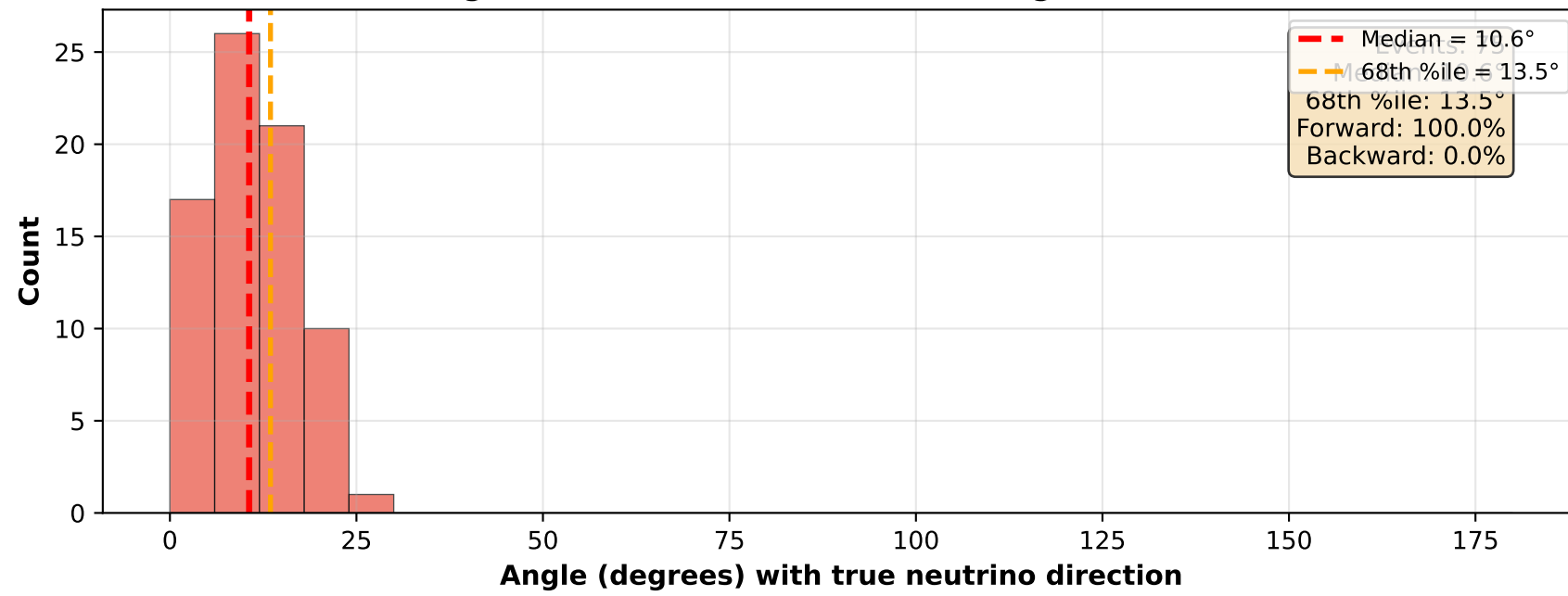
The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.

Input Data Statistics: Files and Clusters per Cat

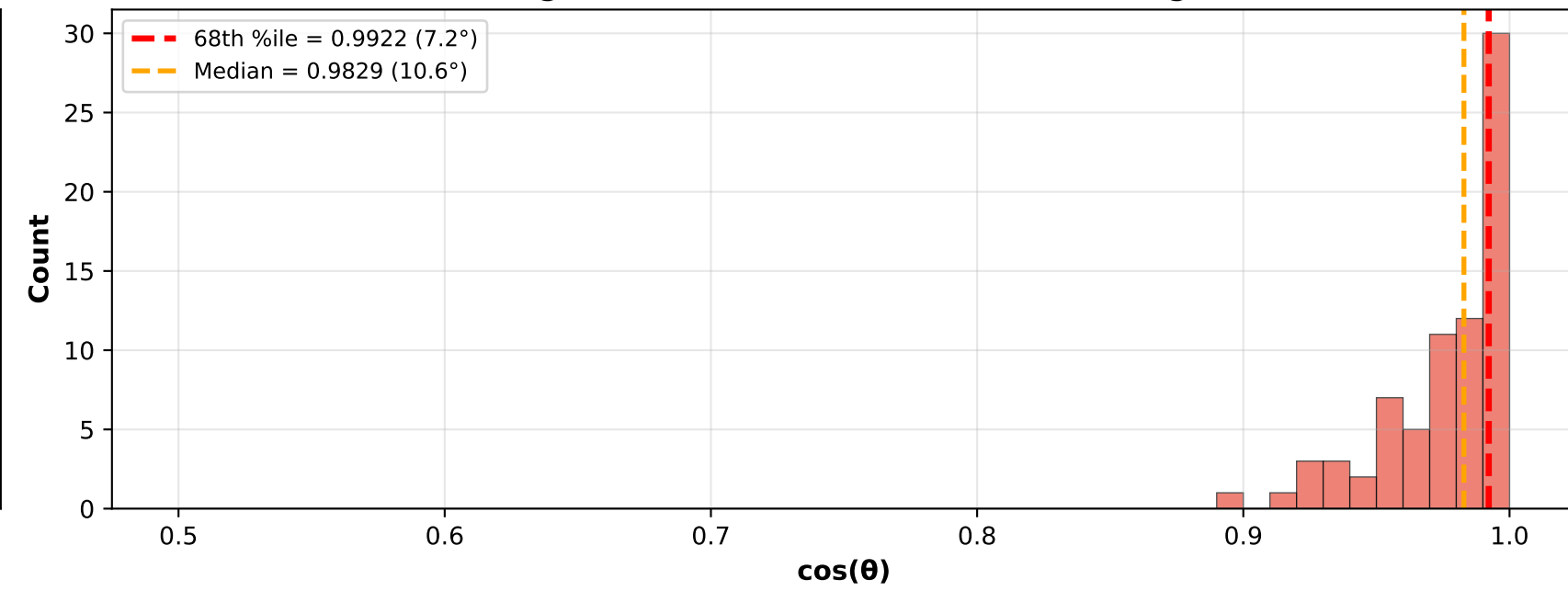


Perfect CT (ES + ED) - Axis-Aligned Analysis ($\pm 30.0^\circ$ selection)

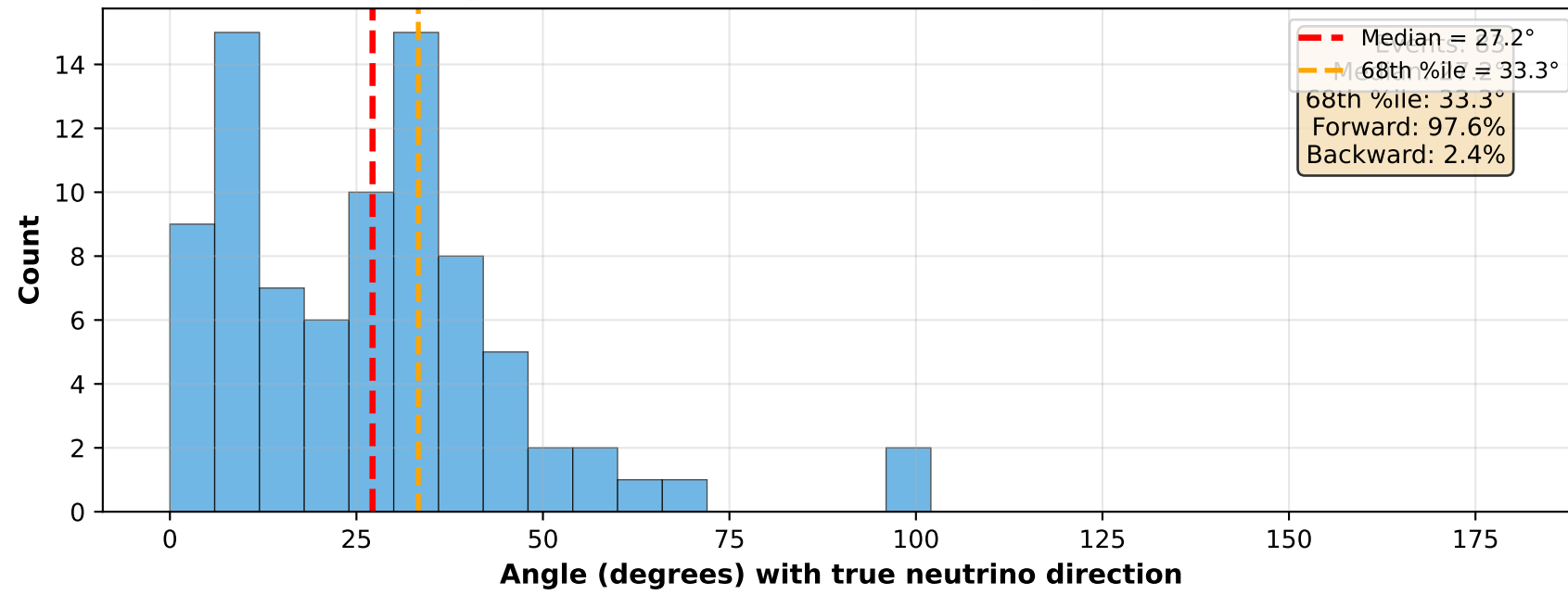
X-aligned (within $\pm 30.0^\circ$ of X-axis): Angle Distribution



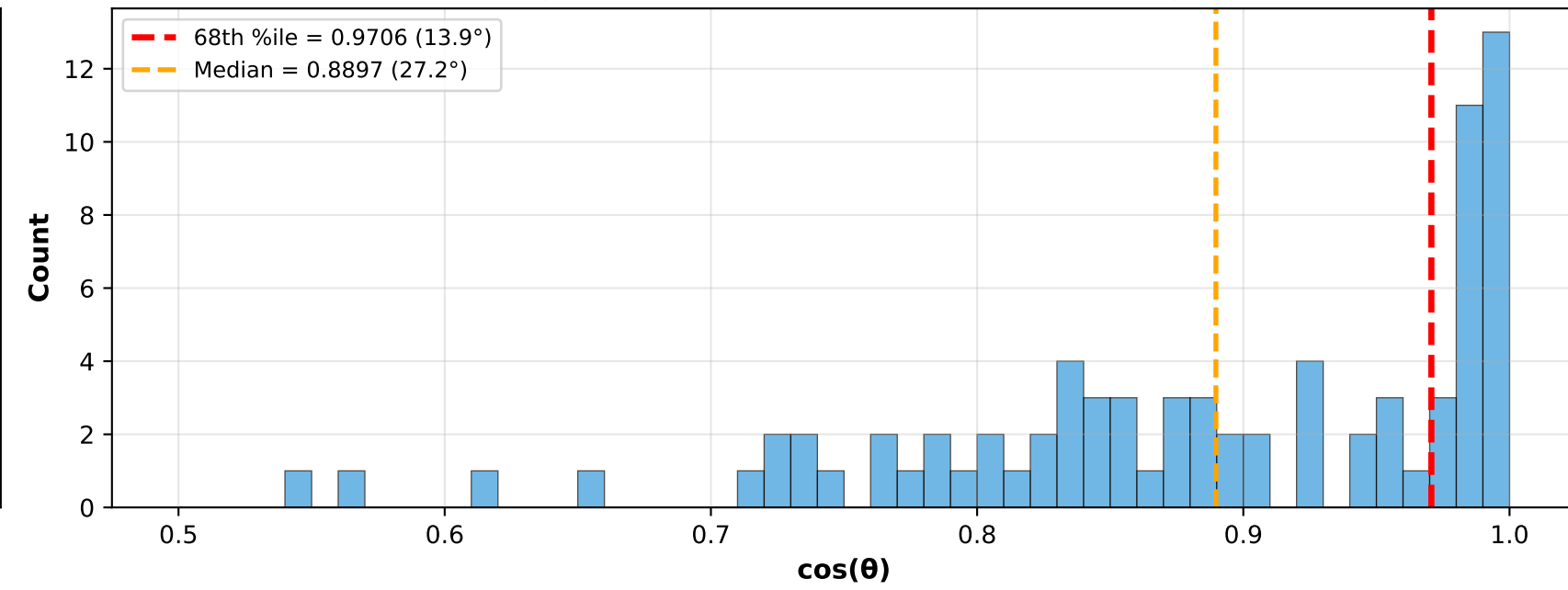
X-aligned: Cosine Distribution [0.5-1.0 range]



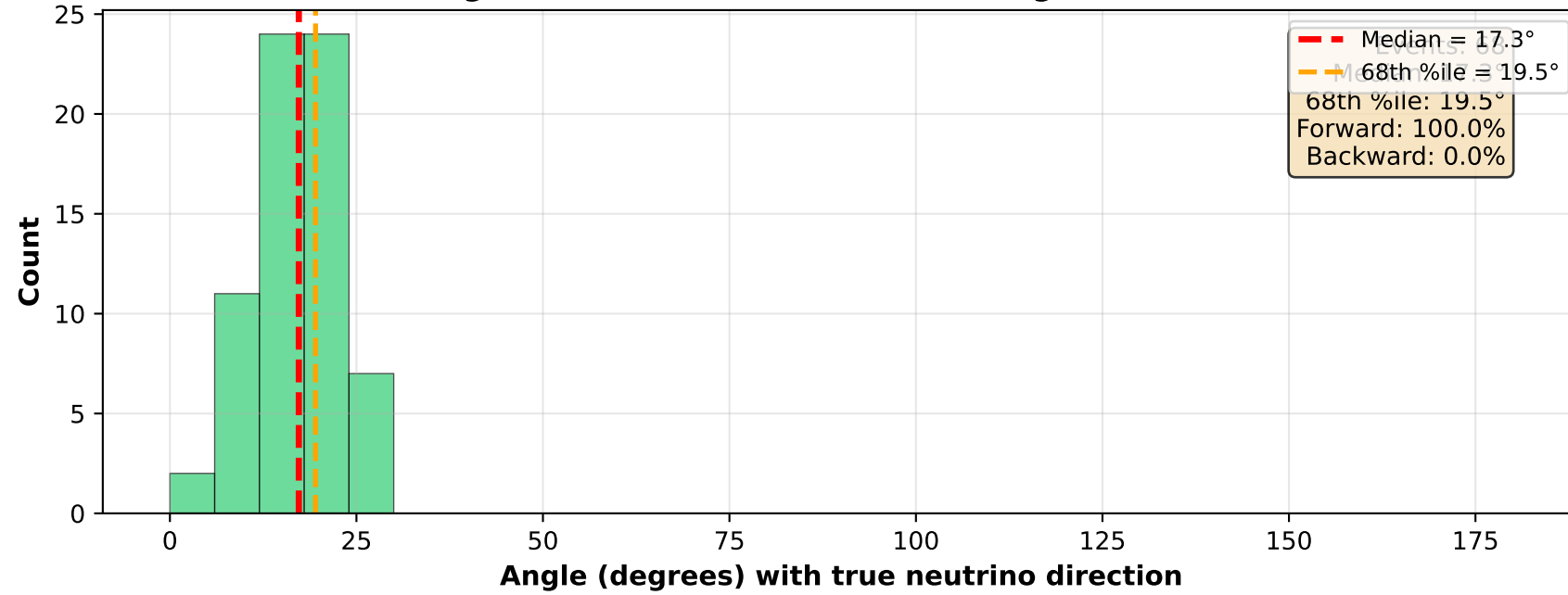
Y-aligned (within $\pm 30.0^\circ$ of Y-axis): Angle Distribution



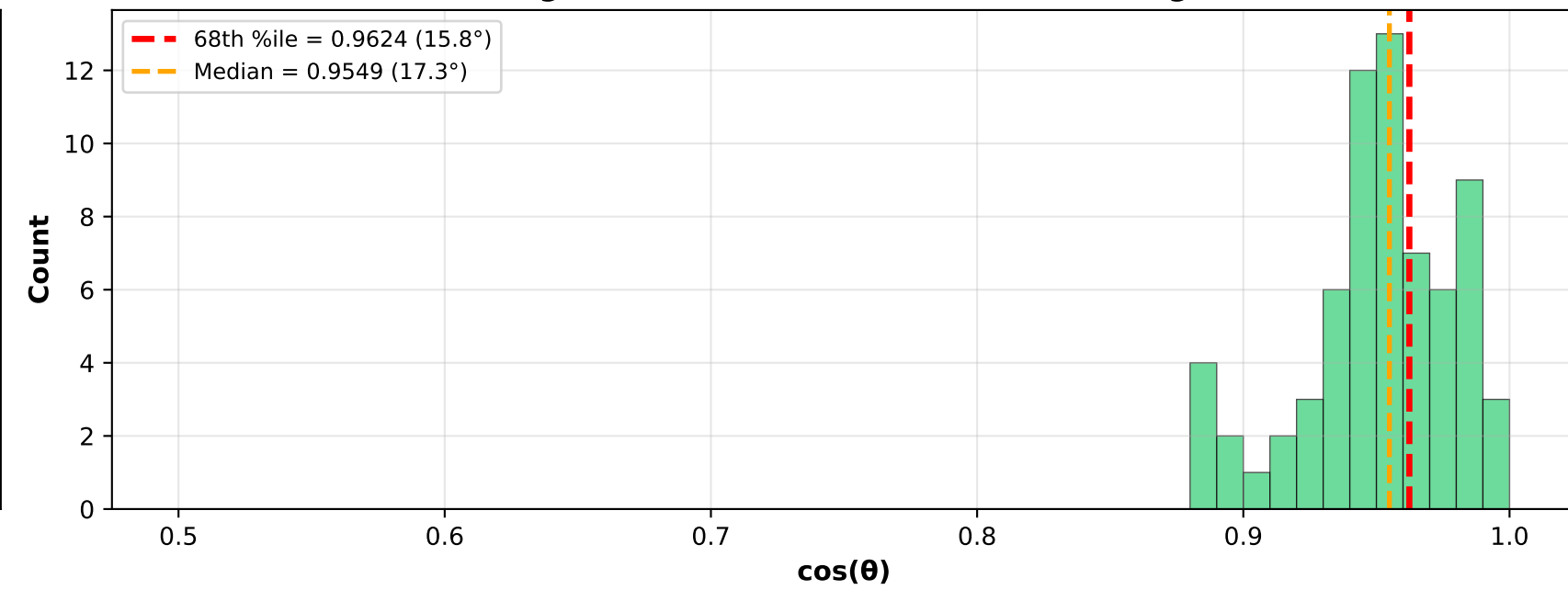
Y-aligned: Cosine Distribution [0.5-1.0 range]



Z-aligned (within $\pm 30.0^\circ$ of Z-axis): Angle Distribution



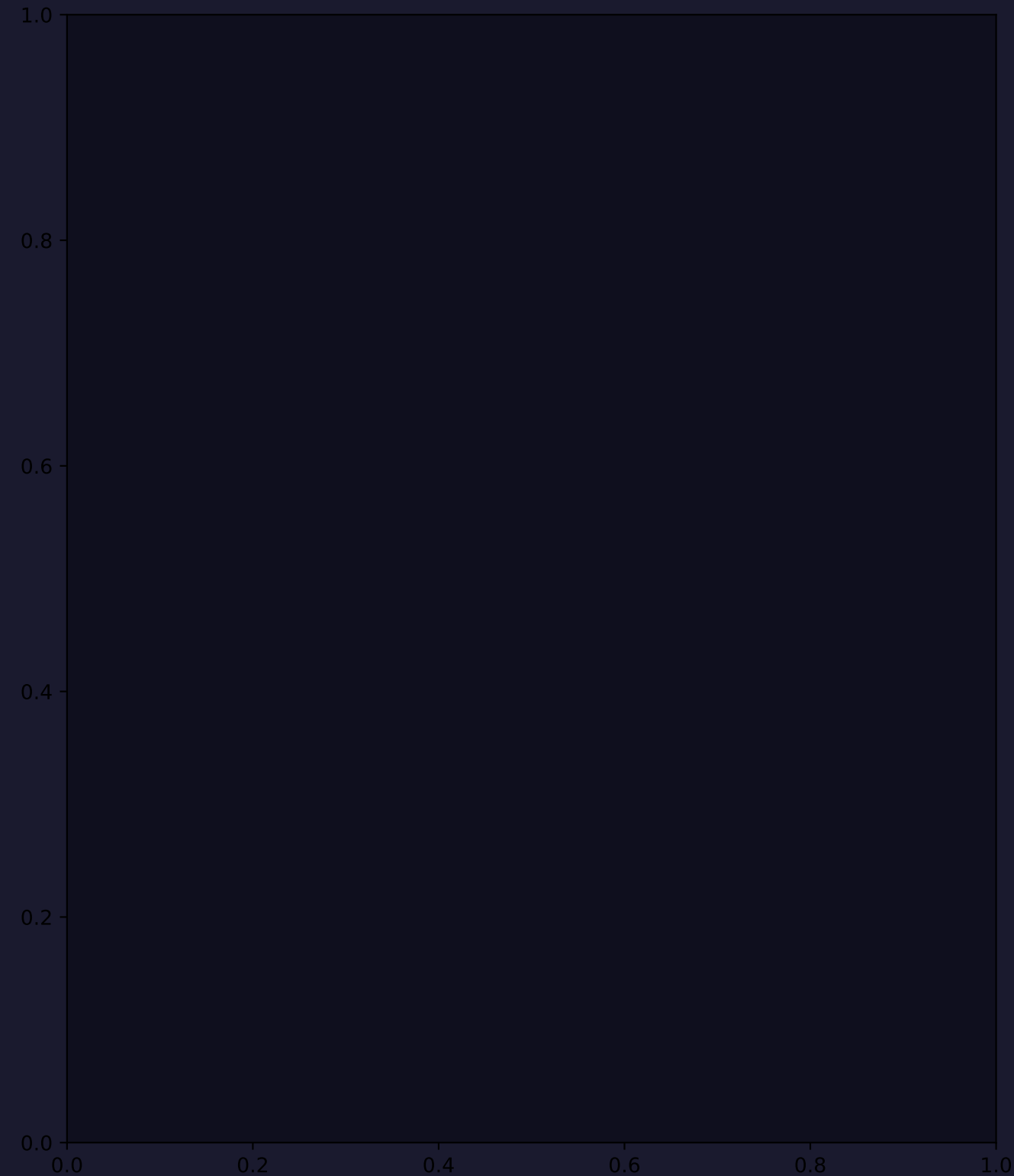
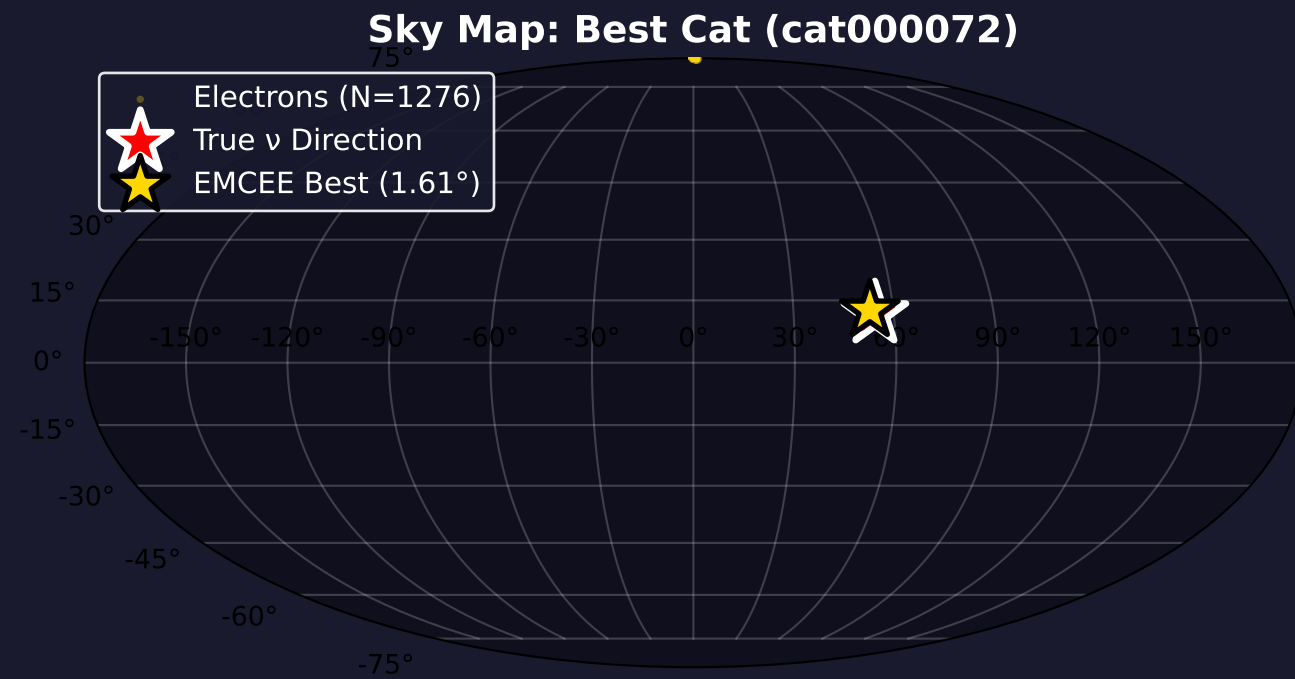
Z-aligned: Cosine Distribution [0.5-1.0 range]



NOTE: Axis alignment is based on TRUE neutrino direction. X-axis performance depends on detector geometry (drift direction, wire orientation). Verify X/Y/Z coordinate definitions match your detector setup.

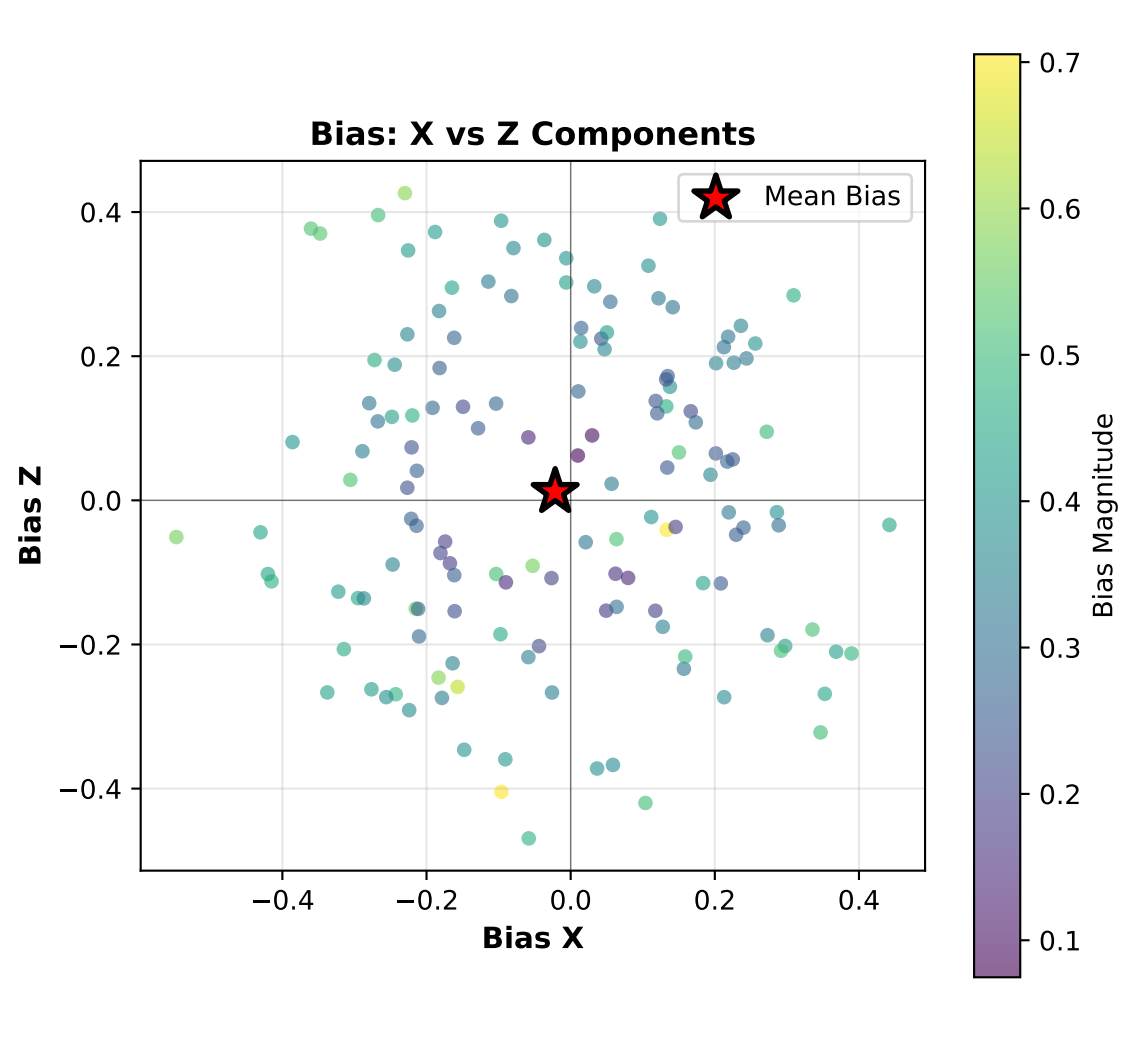
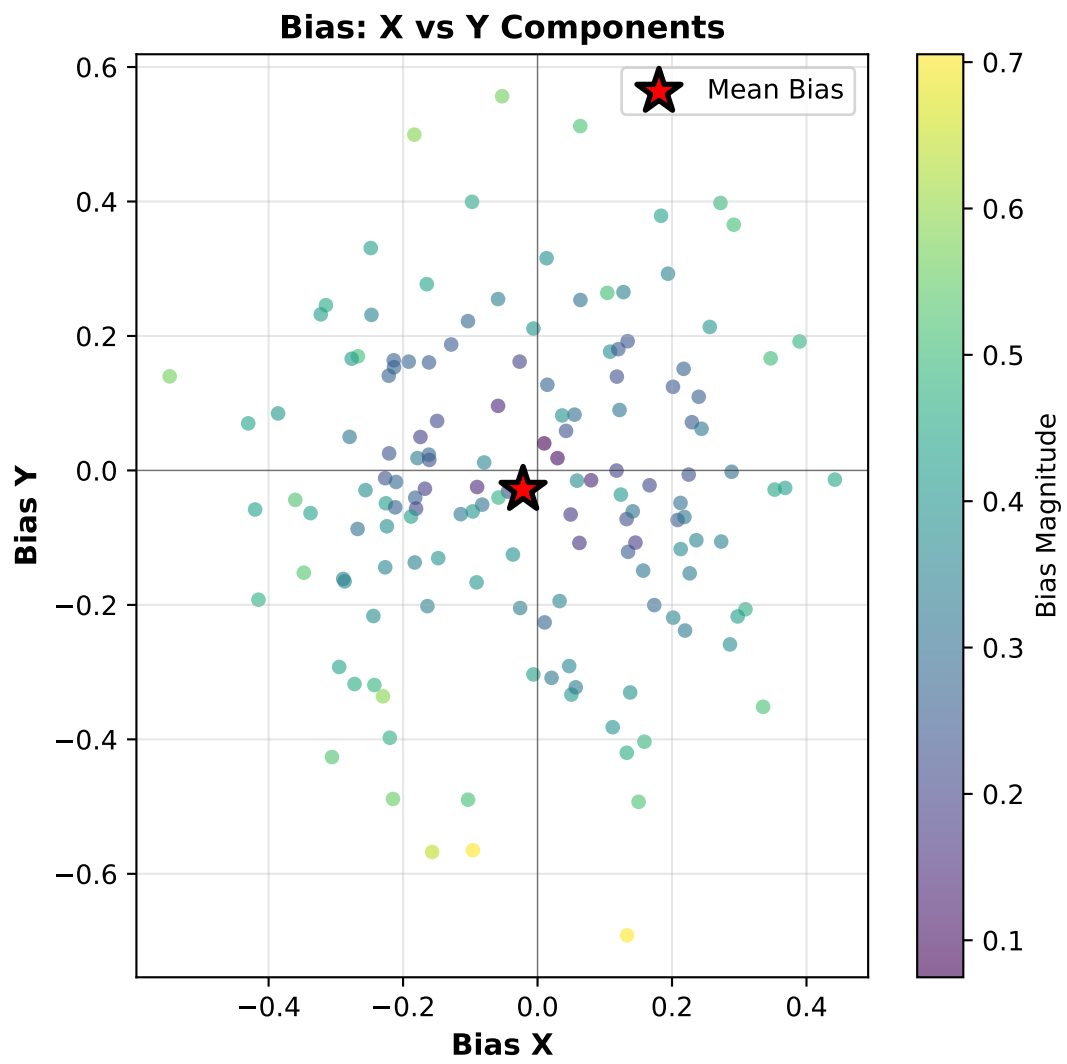
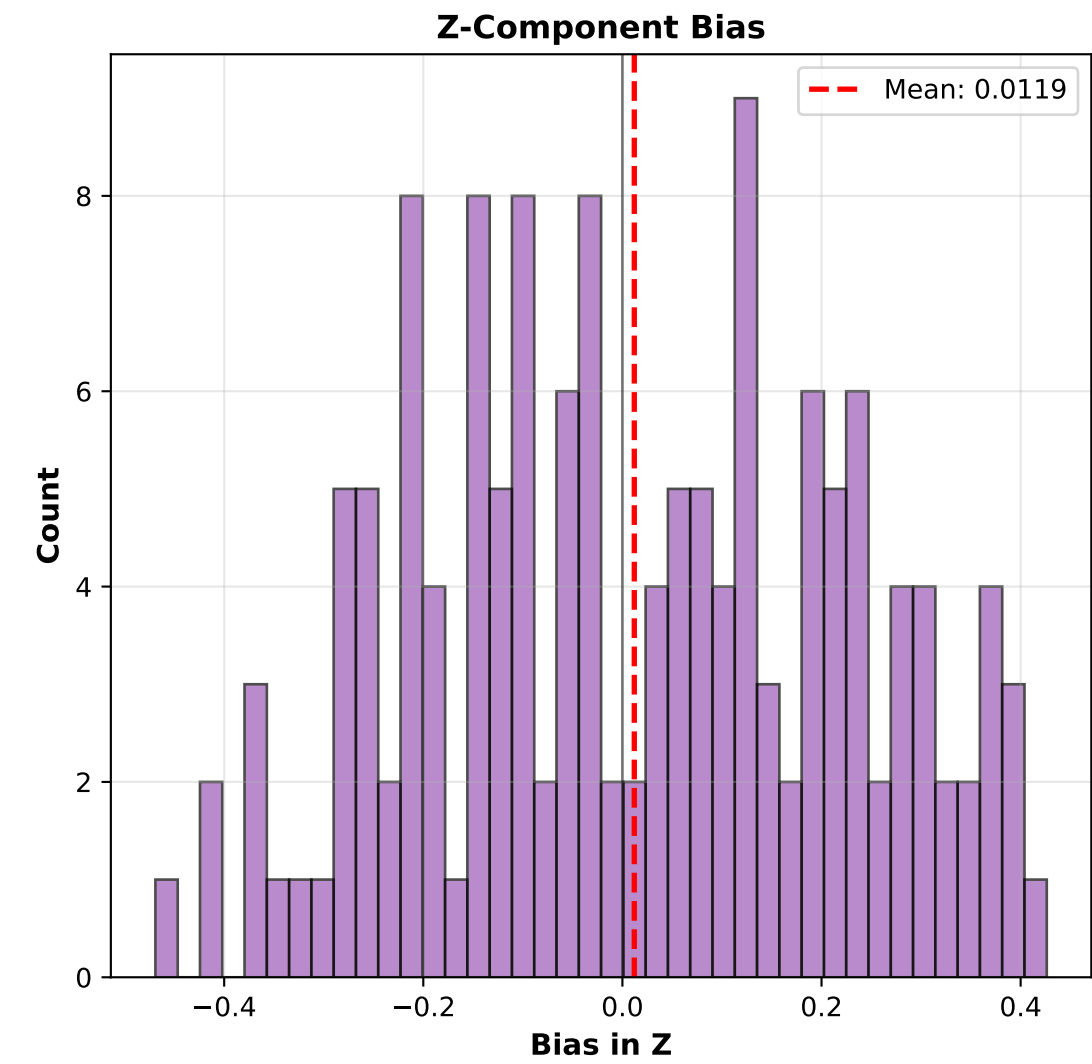
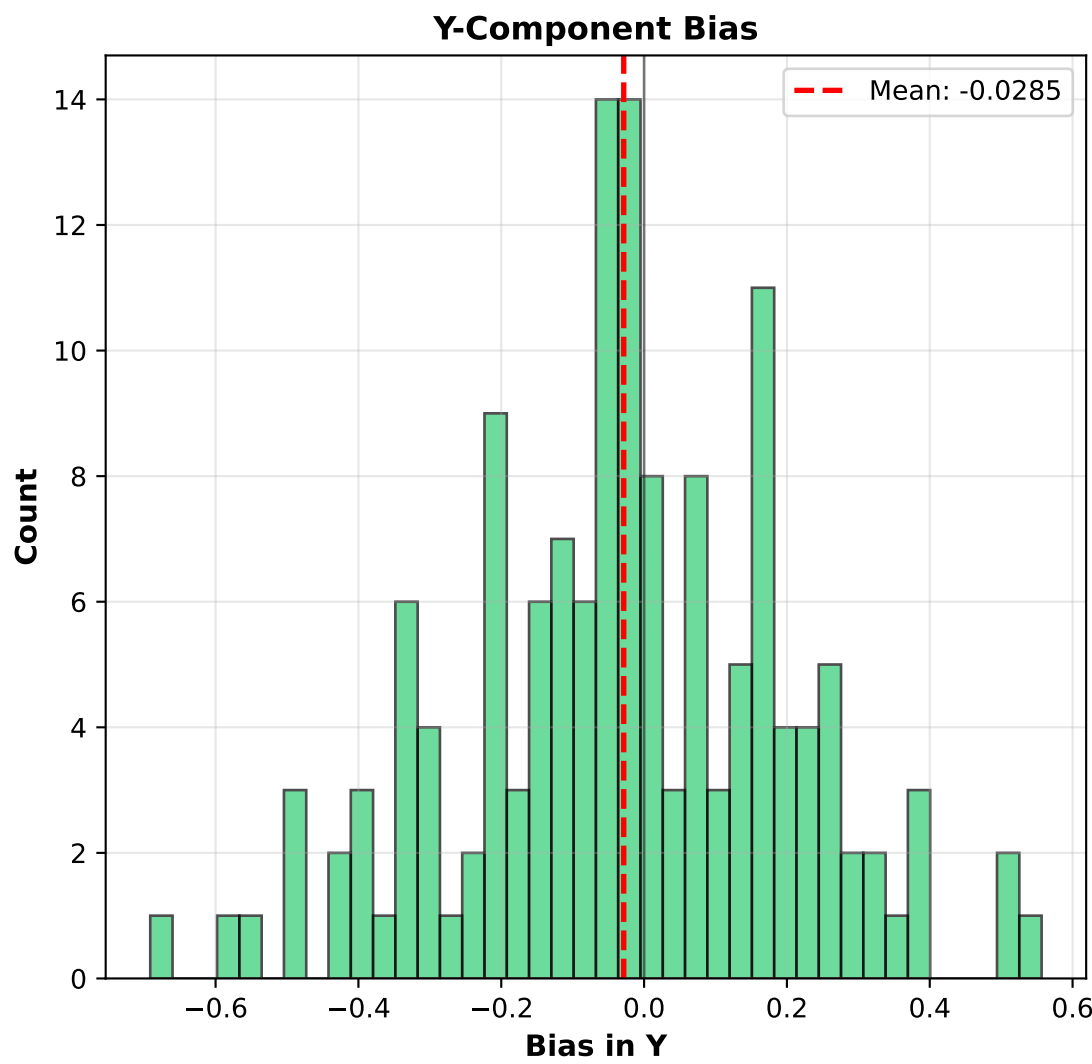
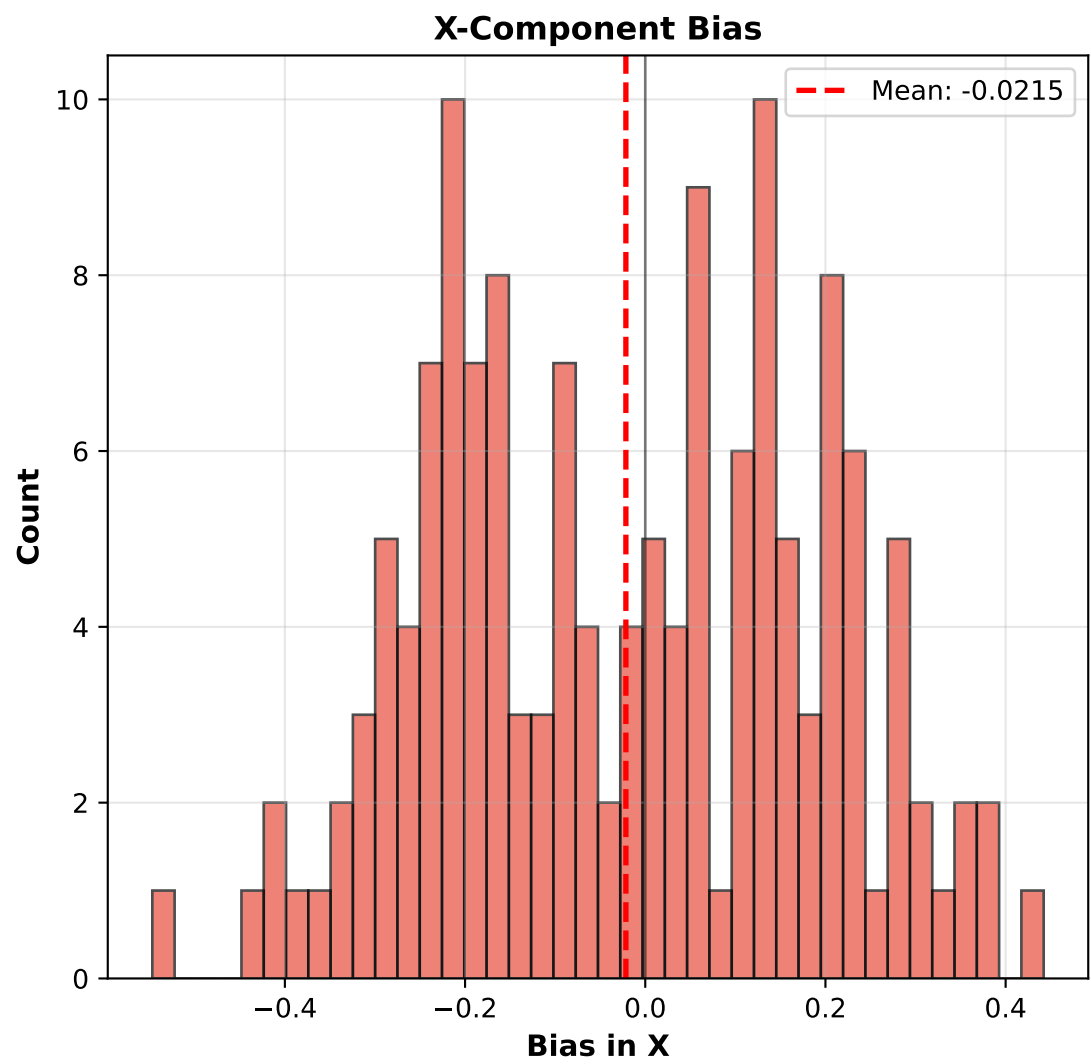
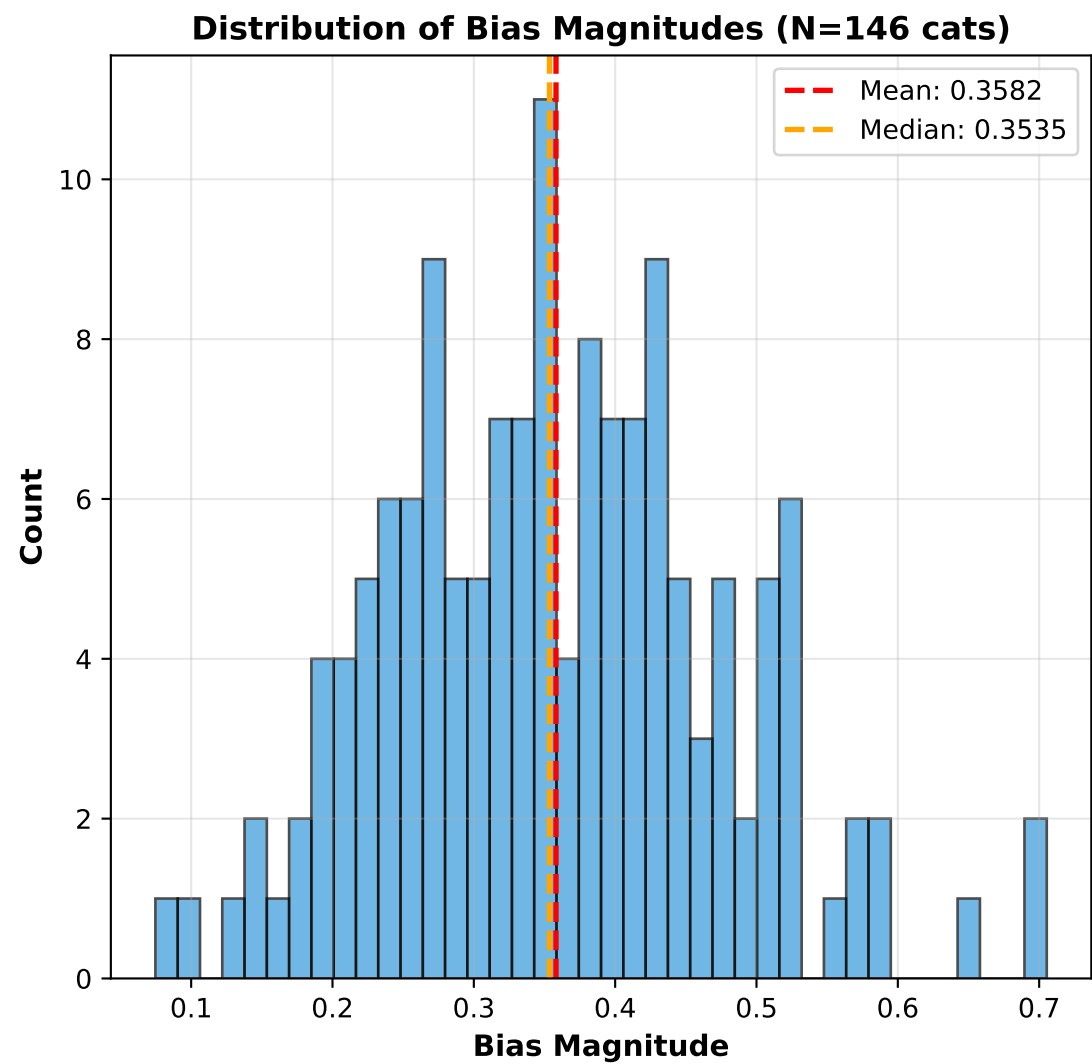
Neutrino Pointing Reconstruction: cat000072 - Perfect CT Scenario

Yellow: Electrons | Red: True Neutrino Direction

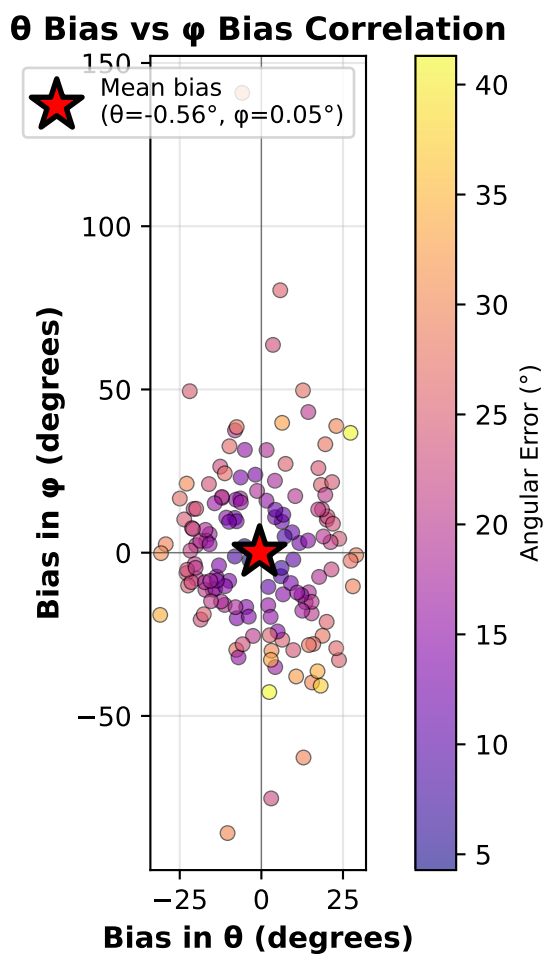
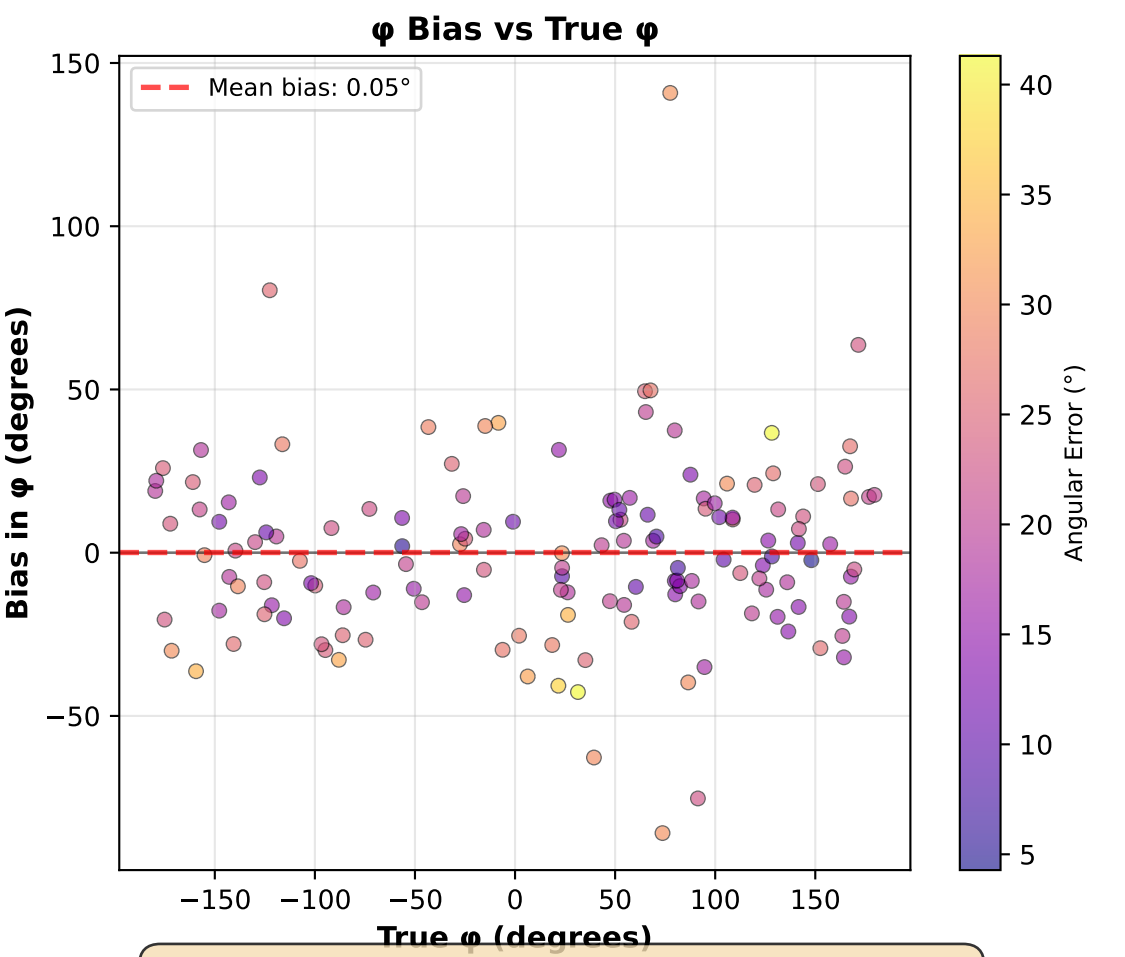
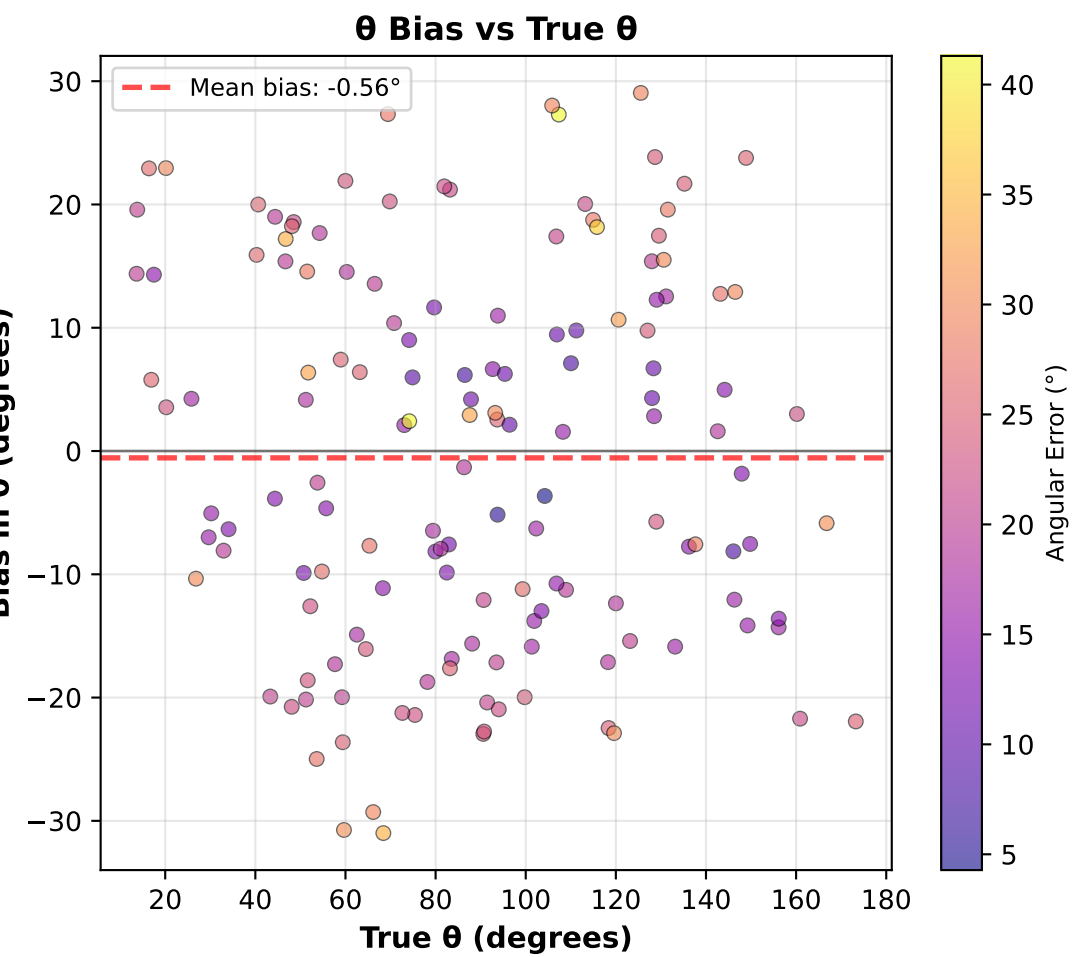
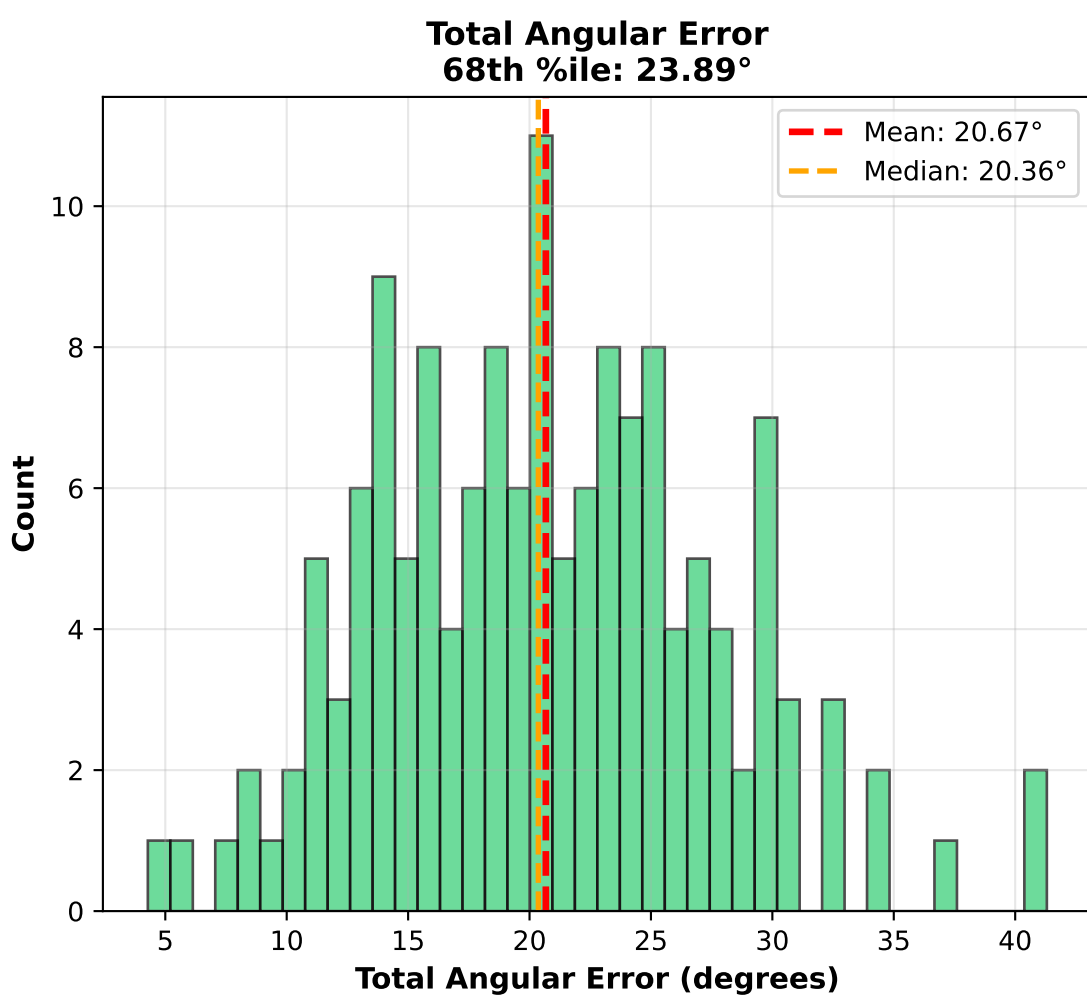
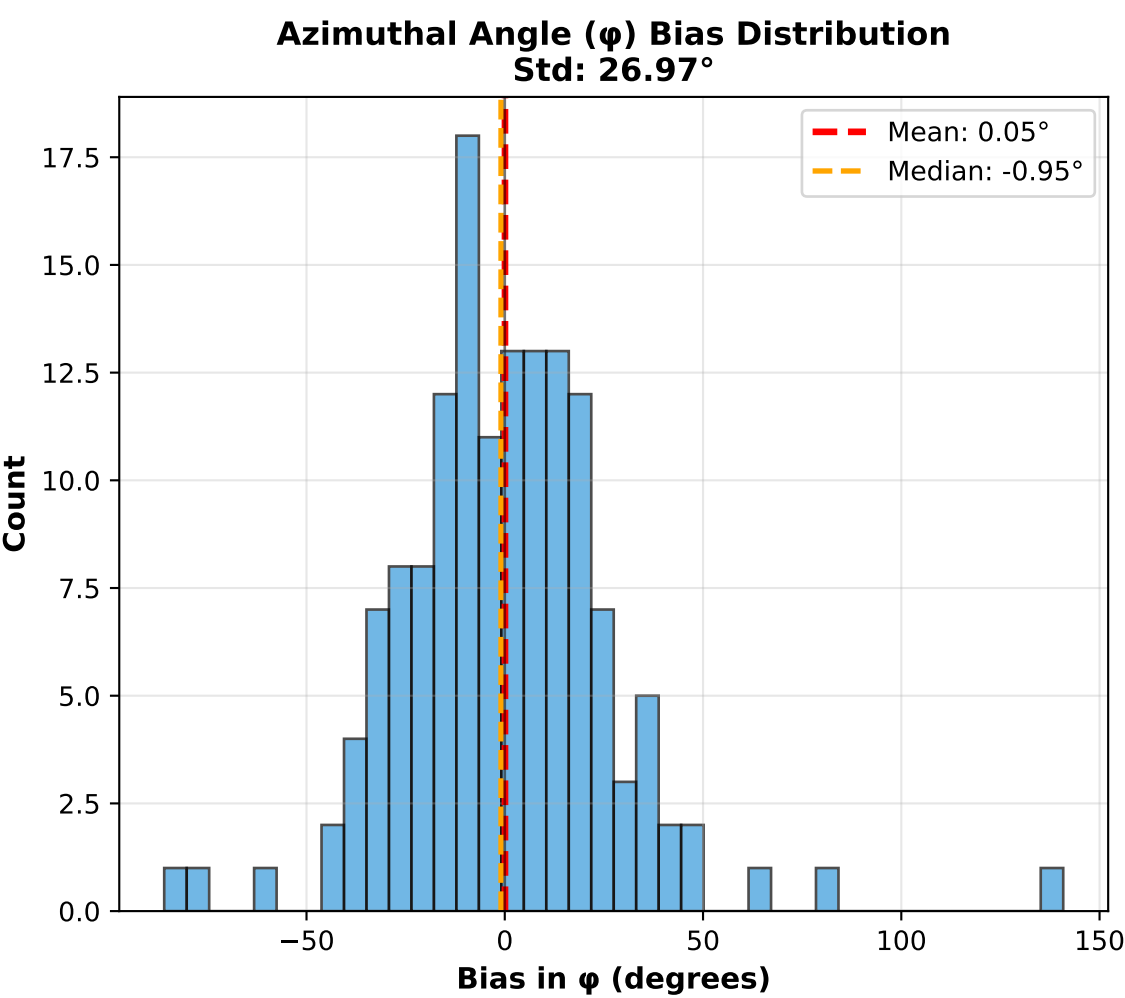
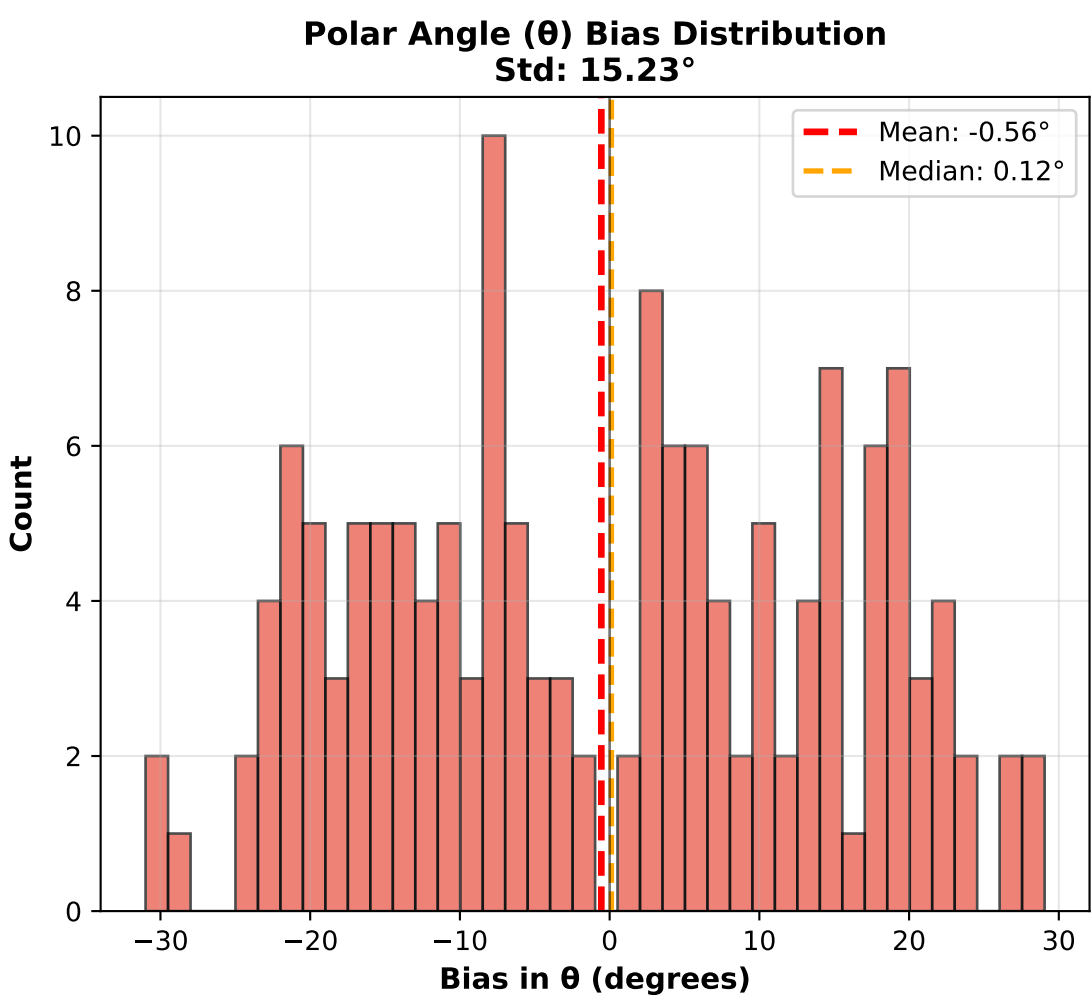


Best cat from EMCEE analysis: cat000072 | Angular Error: 1.61° | 68% Credible Interval (ω_{68}): 0.33° | N electrons: 287

Directional Bias Analysis (Perfect CT) - Cartesian Components
Mean Bias Vector: (-0.0215, -0.0285, 0.0119), Magnitude: 0.0377



Angular Bias Analysis in Spherical Coordinates (Perfect CT)
 θ = polar angle from z-axis [0°, 180°], ϕ = azimuthal angle from x-axis [-180°, 180°]



Angular Bias Statistics (N=146 cats):
θ bias: mean=-0.56°, median=0.12°, std=15.23°
φ bias: mean=0.05°, median=-0.95°, std=26.97°
Total angular error: mean=20.67°, 68th %ile=23.89°